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Research article

Optimizing automated compliance checking with ontology-enhanced natural language processing: Case in the fire safety domain

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ABSTRACT

The fire safety compliance checking (FSCC) plays a crucial role in ensuring the quality of fire engineering design and eliminating inherent fire hazards. It requires an objective and rational interpretation of fire regulations. However, the texts of fire regulations are filled with numerous rules related to spatial limitations, which pose a significant challenge in interpreting them. The current method of interpreting these rules mostly relies on manual translation, which is not efficient. To address this issue, this study proposes an innovative automated framework for interpreting rules by combining ontology technology with natural language processing (NLP). Through the utilization of pre-trained language models (PLMs), concepts and relationships are extracted from sentences, a domain-specific ontology is established, spatial knowledge is transformed into language-agnostic tree structures based on the ontology, and the semantic components of spatial relationships are extracted. The tree structure is then mapped to logical clauses based on semantic consistency, thereby improving the efficiency of interpretation. Experimental results demonstrate that the architecture achieves an ***F1*** score of **86.27** for entity extraction and **81.81** for spatial relationship joint extraction tasks, with an accuracy of **96.26%** in the formalization of logical rules, highlighting its proficiency in automatically interpreting fire spatial rules. This study offers technical support to enhance public understanding of fire safety management and fire prevention predictions, thereby promoting the intelligent management of the building safety environment.

1. Introduction

Raising public awareness of fire safety fosters communities with robust safety environments and is complemented by practical compliance inspections of building fire safety to ensure effective management of the built environment (Daniels et al., 2024) (Santos et al., 2021). Raising awareness promotes building code compliance within the Architecture, Engineering, Construction, and Operations (AECO) industry, while thorough compliance inspections are essential for enforcing these standards (İlal and Günaydın, 2017). Only a combination of public awareness and stringent compliance reviews will ensure a more comprehensive, science-based approach to protecting the community's fire safety environment (Daniels et al., 2024) (Santos et al., 2021) (İlal and Günaydın, 2017).

Fire safety standards delineate a comprehensive suite of obligatory installation parameters for a range of fire protection facilities within the domain of fire safety. Numerous provisions explicitly define the spatial constraints governing the relationship between fire protection facilities and their adjacent environments (Ismail et al., 2023). An illustrative instance of such spatial constraint stipulates that "detectors, when installed on the ceiling of an indoor aisle narrower than 3 m, must be centrally positioned, and the installation gap for spot-type heat detectors must not surpass 10 m" Compliance with these spatial constraints is pivotal for the harmonized spatial integration of fire protection facilities, thereby safeguarding the structural and functional integrity and mitigating fire safety risks attributable to inappropriate spatial arrangements (Xu and Cai, 2018). Presently, the compliance review process predominantly hinges on the experiential acumen of decision-makers, entailing a manual examination of extensive rule data. This approach is laborious, inefficient, and falls short in systematically and uniformly analyzing spatial constraint information within the regulations, owing to the variability inherent in human experience (Zheng et al., 2022). Consequently, there is an urgent need for an automated methodology capable of consistently interpreting the spatial rules embedded within fire regulations text. This will achieve the digital storage of fire safety regulations, optimize the understanding and

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prediction of fire safety management, and promote advanced intelligence in the management of the building safety environment.

A multitude of methodologies have significantly advanced the automation of rule interpretation, exemplified by initiatives like Singapore's CORENET project (Eastman et al., 2009) and the extensively utilized ARC (Automated Rule Checking) application, Solibri Model Checker (SMC) (Greenwood et al., 2010). However, these approaches necessitate manually and expertly hard-coded, rule-based mechanisms, leading to elevated maintenance costs, rigidity in modification, and their frequent characterization as "black box" methods (Nawari, 2019). Recently, the advent of Natural Language Processing (NLP) technology has revolutionized the field by translating human language into formats both comprehensible and executable by computers (Ding et al., 2022). This innovative technology facilitates the extraction of semantic information from textual documents, restructuring it into coherent logical expressions (Xu and Cai, 2021). Additionally, to strengthen NLP-based approaches in rule interpretation tasks, techniques like ontologies, knowledge graphs, and deep learning are employed to explore how large language models (LLMs) can improve knowledge representation, enhancing integration across applications, including openBIM, for a more comprehensive understanding of regulatory texts (Ding et al., 2022) (Chen et al., 2023).

In the realm of fire regulation interpretation, technical challenges include: (1) Understanding and translating the spatial constraints and requirements outlined in fire safety codes into a format that can be processed and comprehended by automated systems. This task involves analyzing textual descriptions within the codes (Kincelova et al., 2020). (2) While some studies have introduced the concept of ontology to elucidate the relationship between concepts, most domain knowledge ontologies are manually created by domain experts, and there are limited ontologies developed for spatial knowledge (Zhou and El-Gohary, 2017). Research on this topic usually divides the construction of ontologies into two parts: named entity recognition (NER) and relation extraction (RE) (Lin et al., 2023). NER is utilized for identifying and uncovering domain concept entities within extensive datasets, whereas RE is employed to algorithmically extract potential relationships among diverse entities (Ligong, 2007). The application of manual methods in NER and RE tasks often leads to an overlook of the interdependencies between entities and their relationships (Yadav and Bethard, 2018) (Zhang et al., 2018). Simplified, while the traditional approach segregates NER and RE tasks, a unified model that simultaneously addresses both proves more effective (Bekoulis et al., 2018), as entity knowledge assists in relationship selection, and relationship understanding aids in entity identification. Consequently, an effective NLP methodology for interpreting fire regulations necessitates the capacity to comprehend spatial semantics from natural language and to simultaneously consider the interplay between entities and their associated spatial relationships. Existing natural language processing methods in the field of architecture have proven successful in semantic-based information extraction by incorporating ontologies (Zhou and El-Gohary, 2017). However, these methods are still limited in the scope of spatial semantics research, despite the development of NLP methods for fire departments in the literature (Zhou et al., 2022), they still face challenges in limited semantic and spatial understanding capabilities.

To address the above-mentioned issues, we introduce a comprehensive framework for interpreting spatial constraint rules. The framework is based on modeling concepts and spatial relationship knowledge within an ontology, encompassing both domain and spatial semantics. It utilizes rule-based NLP technology for the automated interpretation of fire regulations. Enhancements in the framework have been achieved in three key areas: (1) automating the construction of a more precise ontology for the fire spatial domain through a joint extraction model for entity relationships; (2) formalizing spatial knowledge and extracting semantic information related to spatial constraints based on the predefined structure of spatial knowledge; (3) mapping the harvested semantic information to its equivalents in the ontology and transforming it into logical clauses, thereby realizing both spatial semantics and logical formalization. This methodology has been validated through application in specific case studies. Our innovative framework combines ontology technology with NLP to extract and formalize spatial relationships and constraints directly from regulatory texts. This transformation into machine-readable formats is essential not only for developing automated systems that can conduct consistent and accurate compliance checks but also for enhancing operational efficiency in safety measures across building projects. The digitized and specific representation of spatial relationship knowledge can substantially boost public awareness of fire safety knowledge, contribute to public engagement in fire safety management, and improve the intelligence of community building safety and environmental oversight (Ma et al., 2020).

2. Research background

2.1. NLP-based structuring of regulatory provisions in the fire safety domain

Currently, Building Information Modeling (BIM) technology stands as a pivotal core technology within the domain of architectural engineering. The integration of Natural Language Processing (NLP) technology into building information models has emerged as a prominent subject of interest in the field (Ding et al., 2022). At the same time, NLP technology has assumed a significant role in the analysis and scholarly examination of architectural regulations and the dissection of architectural texts (Ma et al., 2022a).

In the study of architectural fire safety regulations, the core purpose of structuring regulations is to convert fire standard specification texts into computationally processable structured data. Nadkarni et al. (Habash, 2010) highlighted that NLP algorithms predominantly employ two effective and widely-utilized methodologies: rule-based and statistical-based methods. The rule-based approach hinges on manually-defined symbol matching rules, offering broad applicability and ease of use, particularly for architectural experts lacking a programming background. Specific examples include the Backus-Naur Form (BNF) method, regular expression syntax, and Prolog language (Wang et al., 2021) (Shadbolt et al., 2006). The statistical-based approach, conversely, automatically constructs "rules" derived from probabilities in training datasets, akin to machine learning algorithms, involving substantial annotated text data. This approach encompasses techniques such as Support Vector Machine (SVM), Hidden Markov Model (HMM), Conditional Random Field (CRF), and Naive Bayes algorithms. While traditional rule-based and probability-based information extraction methods are straightforward to implement, they are constrained in their capacity for feature extraction. Based on this, Luo et al. (Luo and Gong, 2014) employed decision tables and product representation techniques to extract information from standard specifications, simplifying the rules through a comprehensive evaluation methodology. Li et al. (2016) implemented NLP techniques to automatically extract information from standard specifications, leveraging the inherent language structure and critical information elements of the specifications. Zhou et al. (2022) harnessed NLP along with a suite of extraction techniques, including ontology-based pattern matching, sequence dependency-based, and cascaded extraction, to derive information from intricate energy-saving codes. Wei et al. (2019) conducted structured analyses of 16 prevalent Chinese fire design specifications using context-free grammar, achieving structured rule expressions for six distinct specification types. Zhang et al. (Zhang and Gu, 2018) studied the method of automatically extracting domain knowledge from specification texts based on natural language processing in the BIM-Checker tool but did not involve automation and intelligent integration at the spatial semantic level. Zhang et al. (Zhang, 2014) discerned that the efficacy of word segmentation can be substantially enhanced through the creation of a domain-specific dictionary. Consequently, they amalgamated machine learning methods with user dictionaries in the training of word segmentation, thereby augmenting the cross-domain adaptability of Chinese word segmentation. This approach relies on the quality of the domain-specific lexicon; incomplete or poorly managed lexicons can result in suboptimal segmentation. Furthermore, the study does not evaluate performance in data-scarce domains, which limits its applicability in various real-world scenarios. Song et al. (Song, 2021) utilized knowledge graphs to enhance the application of safety knowledge within the building safety domain. They proposed a named entity recognition model tailored for the building safety field, which, through experimental validation, demonstrated superior recognition effects compared to traditional methods. However, it focuses exclusively on named entities and omits the relationships among them within the knowledge graph, potentially restricting the model's ability to capture the complete context and interconnections essential for a comprehensive security assessment.

2.2. Joint entity-relation extraction for domain-specific ontology construction

Entity-relation extraction represents a fundamental task and a critical aspect within the realms of information extraction, knowledge graphs, and Natural Language Processing (NLP). It is also a crucial step in the automated construction of ontologies, where knowledge is represented using a universal data model called RDF. This model is based on labeled directed edges between nodes, forming triples. All knowledge in the objective world can be articulated through structured triples [S, P, O], with S, P, and O denoting the subject, predicate, and object, respectively, correlating to the entities and the interrelations between entity1 and entity2 (Ji et al., 2022). The construction of ontologies is typically done in three ways: manual construction, semi-automated construction, and automated construction. With the growing recognition of the effectiveness of automated construction methods, it has become a hot research topic. At present, researchers primarily utilize supervised learning methods for automated ontology construction, focusing on named entity recognition (NER) and relation classification (RC) informed by expert knowledge. Kethavarapu and other scholars have used keyword-based and value-based extraction methods to integrate log file data and convert it into an OWL file, thereby achieving automatic ontology generation (Kethavarapu and Saraswathi, 2017). Conversely, Zhao and peers (Zhao et al., 2015) proposed a novel structural word extraction approach based on multi-label learning models and associated label propagation, aimed at enhancing the accuracy of ontology relation recognition and optimizing ontology construction. Zheng et al. (2019) employed the Latent Dirichlet Allocation (LDA) topic model for concept extraction within the realm of user-generated content, validating the efficacy of the LDA algorithm in this context. Christou et al. (Christou and Tsoumakas, 2021) proposed a joint label embedding method using the Bidirectional Encoder Representations from Transformers (BERT) model and graph attention mechanism, which improved the effectiveness of entity-relation extraction through remote supervision. Aleksandra and others evaluated various relation extraction models, including the BERT model, on the Text Analysis Conference Relation Extraction (TACRED) dataset, providing a standard benchmark for evaluating entity-relation extraction performance (Alt et al., 2020). However, most studies utilize publicly available datasets for training and are not specific to the fire safety domain.

2.3. Knowledge gaps

Although many studies have made significant contributions to the automated interpretation of fire regulations, there are still some limitations in the following two aspects.

Firstly, the current methods of natural language processing (NLP) are limited to processing firefighting texts at a coarse-grained level, predominantly conducting analyses at the sentence level, without adequately accounting for the intricate relationships between words and their contextual backdrop (Luo and Gong, 2014) (Li et al., 2016). While some scholars have integrated machine learning techniques into this process (Habash, 2010), it still requires a large amount of manual annotation work to achieve better accuracy, which is time-consuming and labor-intensive (Wei et al., 2019) (Zhang and Gu, 2018). In addition, firefighting regulations are replete with modal verbs, including terms like "should" and "not suitable". In the process of structuring the text, these deontic modal logics should be fully considered to improve the generalization of logical representations.

Second, entity-relation extraction is a typical application of information extraction. Certain studies have successfully accomplished automatic extraction of entity-relation and the automated construction of ontologies. However, most of these studies separate NER and RC tasks (Zhou et al., 2022). There exists a significant correlation between entities and relations. In other words, entities and relations influence each other. A single entity can correspond to multiple relations, and conversely, a single relation may associate with various entities. An illustrative instance is the directive: "When installing the control center monitoring equipment, the bottom edge should be situated 100 mm–200 mm above the ground (floor)." In this scenario, the fire entity "control center monitoring equipment" manifests two types of relations: a spatial relation with the ground and a numerical constraint relation of "100 mm–200 mm". Disregarding the dependency between entities and relations can lead to a disruption in their semantic correlation. Consequently, it is imperative to concurrently consider both entity extraction (NER) and relation classification (RC) tasks to amplify the informational interplay between entities and relations.

3. Method

Our research focuses on the automation of space constraint rules interpretation methods, with a foundation in the fire and geographic spatial information industries. It is essential for this study to explore the respective characteristics and inherent connections of the geographic spatial attributes of fire equipment. Consequently, we integrate an entity-relationship joint extraction model for ontology construction, comprehensively considering the interplay between entities and their relationships. Based on this, the Context-Free Grammar (CFG) algorithm is introduced to formalize spatial knowledge and combined with the expressive method of deontic modality unique to normative articles, the result is transformed into logical rule clauses. Fig. 1 illustrates the research process, which consists of four distinct steps. It is important to note that we have highlighted the key aspects of each step in different colors, and these highlights will be elaborated upon in the following sections.

Step 1. Preprocessing (Fig. 1-Phase #1)

Analyzing the statements and clauses of fire safety regulations to obtain the content of clauses with spatial constraints, defining the different categories and spatial relationships of fire components, and preprocessing these sentences by segmenting them to facilitate subsequent work. The process described above is primarily conducted manually.

Step 2. Automated construction of ontology (Fig. 1-Phase #2)

In this phase, deep learning neural networks are employed to propose a novel joint extraction model, termed Fire-GeosBERT. The model is tested and improved for specific downstream tasks, enabling joint extraction of entity-space relationships and automated construction of domain ontology, clarifying spatial relationships between concepts.

Step 3. Space grammar parsing (Fig. 1-Phase #3)

This step utilizes CFG to parse a single sentence into a syntax tree, and along the hierarchical structure of the syntax tree, extracts spatial cognitive semantic elements. By combining with ontology, the spatial relationship triplets are formalized, providing a reliable data foundation

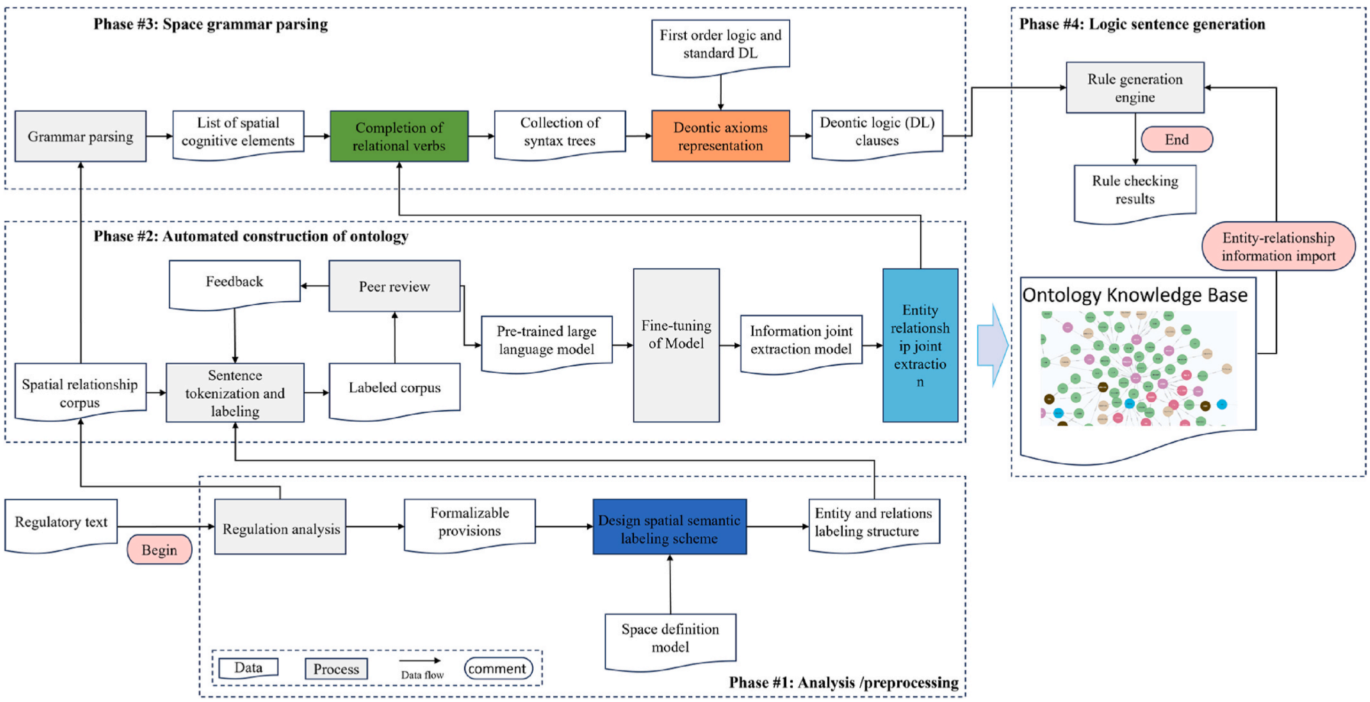

Fig. 1. Interpretation framework for fire safety regulations.

for the generation of logical rules in the subsequent process. This step automates the formalization of fire safety knowledge.

Step 4. Logic sentence generation (Fig. 1-Phase #4)

In this step, the process begins by categorizing the deontic modalities of fire safety regulations, followed by transforming the spatial relationship constraints depicted in the tree structure into logical clauses that embody these deontic modalities. This transformation aids in facilitating rule checks in specific scenarios subsequently. This step implements an automated translation of deontic logic and establishes the foundation for subsequent automated rule checking.

To elucidate the proposed framework more effectively for interpreting fire space rules, this study utilizes the 2018 edition of the "Code for Fire Protection Design of Building GB50016-2014 (subsequently referred to as the "Construction Regulations")" (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2020) as the data source, pinpointing pertinent domain-specific spatial concepts. The dataset pertaining to fire safety regulations encompasses both guiding specifications and detailed executive specification information. Geographic spatial attribute data is primarily derived from the normative textual descriptions detailing the spatial positioning and topological relationships of fire equipment.

3.1. Analysis and preprocessing of fire safety regulations

This section provides a detailed examination of the key processes involved in analyzing and preprocessing fire safety regulations. This phase serves as the foundation of our study, involving a meticulous breakdown and preparation of regulatory texts, which are crucial for the subsequent automated processing and interpretation. Initially, we analyzed the normative and spatial constraints specified in the regulations. Subsequently, we applied preprocessing techniques such as sentence segmentation and syntactic tagging to deconstruct complex legal texts, preparing them as input for subsequent models. These initial steps are essential to ensure the accuracy and efficiency of our ontology-based rule interpretation framework.

3.1.1. Regulation analysis

Based on the provisions in the "Construction Regulations", the content is categorizable into four distinct types of constraint categories: attribute value constraints, relationship constraints, normative constraints, and comprehensive constraints (Liu, 2022). Table 1 shows the specific content of these four types of provisions. Predicated on these four categories, the provisions can be further classified according to whether they can be formalized. Normative provisions can be divided into two categories: formalizable provisions and non-formalizable provisions. Formalizable provisions are characterized by their clearly defined specified values. The review process entails assessing whether the actual parameters of a construction project align with the specified value range, thus determining if they conform to the normative

Table 1
Four examples of types of articles are as follows.

Clause Content	Clause Category	Whether Formalized
5.1.4 The vertical distance from the axis of the linear beam fire detector to the ceiling should be about 0.3 m–1.0 m.	Attribute value constraint category	**True**
5.4.13 Diesel generator rooms installed in civil buildings should not be located on the upper or lower floors, or adjacent to densely populated areas.	relationship constraints category	**True**
5.3.5 The wall of the fire compartment should be a fireproof partition with a fire resistance limit of not less than 3.00 h, and it should comply with the provisions of Article 6.4.13 of this specification.	Normative constraints category	True
5.2.1 In the overall layout of the site, it is essential to carefully determine the placement of the buildings, the fire separation distances, the fire lanes, and the fire water sources. It is not advisable to situate the buildings near Class A and B factories (warehouses), Class A, B, and C liquid storage tanks, flammable gas storage tanks, and combustible material storage areas.	Comprehensive constraints category	False

requirements, as exemplified by provisions in the attribute value constraint and relationship constraint categories. In the context of the third category of normative constraints, the formalization process is more complex, as it requires validation in conjunction with simulation software. Non-formalizable provisions are distinguished by the absence of clearly specified values. Typically, they necessitate subjective evaluations by industry experts or professionals to assess the construction project. The review results cannot be clearly determined, as clause 5.2.1. Due to the characteristics of normative clauses mentioned earlier, we focus on summarizing the normative logic expressions within the categories of attribute value constraints and relationship constraints, and have manually collected the corresponding statutory provisions. In contrast, performance-based rules necessitating validation through simulation software (e.g., Section 5.3.5) demand more comprehensive analysis and constitute the focus of ongoing research, the findings of which will be shared as an extension of the present work.

3.1.2. Preprocessing

The purpose of preprocessing is to segment individual sentences for subsequent work. The preprocessing process includes sentence segmentation, tokenization, hyphen removal, and part-of-speech tagging. The formal characteristics of fire safety regulations writing have been summarized. The existence of these formal characteristics makes it difficult to structure the fire safety regulation articles according to the spatial categories expected in this study. Therefore, it is necessary to preprocess the fire safety regulations to a certain extent.

Due to the inclusion of normative provisions expressed in tabular form in the fire safety regulations documents, considering the relatively small number of tables compared to the number of provisions, the tabular data is manually combined with the content of the table headers according to natural language conventions to supplement and form provisions that possess spatial semantic relationships (Salama and El-Gohary, 2016).

3.2. Definition of spatial knowledge semantic elements

The research specifically focuses on the attribute constraint class and relationship constraint class as its primary subjects. Firstly, rule checking involves the process of locating and identifying objects within the hierarchical structure of the BIM model, followed by assessing these objects in accordance with the stipulated requirements (Zhou et al., 2022). This process can be represented by a tree structure, where some nodes represent the objects being checked, and these nodes are linked to requirements, which contain comparative relationships or existence relationships and the required constraints or conditions. Secondly, the perspective of this study focuses on fire code specifications with spatial constraint attributes. Therefore, it is necessary to assign several types of tags to words or phrases related to fire in the preprocessed sentences to indicate their spatial knowledge semantics. In the domain of fire safety, fire equipment primarily includes fire alarm controllers, fire hydrants, point-type smoke detectors, fire alarm bells, among others. Knowledge of spatial relationships, which includes an understanding of the location and interaction of different building elements and fire safety components in a building layout, is critical to ensuring compliance with fire safety codes. Two types of relationships are involved: topological and positional (Wang et al., 2021). Borrmann and Rank provided a semantic definition of topological relationships in 3D space, including touch, overlap, non-intersect, and equality, by using the 9-intersection model (Borrmann and Rank, 2009). Regarding positional relationships, as per the spatial pattern delineated by Kordjamshidi et al. (2017), six specific types of relative directions are identified: left, right, front, back, up, and down. Position information is invariably described through a synthesis of direction and distance. Based on the above basis, we define eight entity types that can represent the field of fire and six spatial relationship knowledge representation methods, as shown in Table 2.

Table 2
Eight entity semantic representation elements and six relationship semantic representation elements were proposed.

label	Definition
objDoor	Parent node, indicating a collection of door types in the field of architectural fire safety, such as “Fire Doors” and “Safety Doors".
objWall	Parent node, indicating a collection of wall types in the field of building fire protection, such as “Firewalls,” “Load-bearing Walls"
objFPS	Parent node, indicating fire spaces in the field of building fire protection, such as “Fire Pump Rooms” and “Safe Rooms".
objFA	Parent node, indicating a collection of fire lane types, such as “Fire Lanes".
objFE	Parent node, indicating a collection of fire equipment types, such as “Hydrants”, “Fire Alarm Controllers”, “Fire Detectors"
objFC	The sub-nodes indicating firefighting equipment, which constitute the integral components of the firefighting field, are exemplified by “Hydrant Buttons” and “Manual Fire Alarm”. Buttons”, “Fire Phone Jack Locations”
objFP	Parent node, indicating the basic composition element with the floor as a single entity, such as “Ground (Floor)
objSCV	Parent node, indicating the specific numerical value of spatial constraints.
RAbove	Relationship node, indicating A is located above B.
RInterlinked	Relationship node, indicating the contact between element A and element B.
RSeparation	Relationship node, indicating that element A and element B do not come into contact and are separated by a distance.
RIntersection	Relationship node, indicating that element A and element B share a portion.
RContain	Relationship node, indicating that element A is a component of element B.
RSpatial	Relationship node, indicating the numerical constraint between element A or element B and object SCV.

(1) For clause of relational constraints, such as “Fire alarm, sprinkler light alarm, gas fire extinguishing system manual and automatic control status display device should be installed by wall hanging method.” It means that fire protection facilities should be connected to the wall. Therefore, this rule comprises three components: physical objects (such as fire alarm, sprinkler light alarm, gas fire extinguishing system manual and automatic control status display device), relational objects (wall), and the relational link (connection). They are correspondingly defined as “objFE,” “objWall,” and “RInterlinked."

(2) For the attribute constraint clause, exemplified by “The vertical distance from the axis of the linear beam smoke fire detector to the ceiling should range between 0.3 m and 1.0 m,” the most prominent feature is the inclusion of spatial attribute values. This clause comprises three elements: the entity object (axis of the linear beam smoke fire detector), the constraint object (0.3 m–1.0 m), and the relational link (spatial numerical constraint). They are correspondingly defined as “objFE”, “objSCV”, and “RSpatial".

3.3. Semantic labeling scheme

The aim of semantic labeling is to enhance textual data by incorporating semantic insights, resulting in a notable improvement in the machine’s ability to read and interpret such data (Liu et al., 2017). This procedure transforms natural language, which is often ambiguous and has multiple meanings, into a more organized and clearer format. As a result, it facilitates the creation of ontology knowledge bases that can be built upon. Given the vital role that this process plays in the effective implementation of spatial relation rule interpretation architecture, our proposed labeling scheme has been developed based on a logical framework.

Considering the crucial importance of considering relationships in this study, particularly in the process of extracting joint entity relationships, our research puts forth a novel semantic annotation scheme. Distinguishing itself from the conventional BIO annotation scheme solely aiding entity recognition, our proposed scheme offers a structured format for annotating entities and relationships, thereby serving as an invaluable resource for constructing ontology knowledge bases or tackling intricate relationship extraction tasks. This annotation scheme not only labels entities in the text but also provides comprehensive descriptions of the relationships between said entities. The effectiveness and utility of this format in our research are elaborated upon in section 4.2.

3.4. Automated ontology construction

The essence of automating ontology involves the automation of Named Entity Recognition (NER) and Relation Extraction (RE) tasks, followed by the construction of triples. Recently, a significant development in Natural Language Processing has been the advent of a large-scale pre-training language model (PLM) known as the BERT model, which has demonstrated commendable performance across various NLP tasks. BERT achieves an understanding of linguistic nuances by pre-training a deep bidirectional transformer network on extensive text corpora. This training process yields more nuanced word representations by leveraging context from both preceding and succeeding tokens within sentences, thereby enabling richer contextual interpretations of text (Devlin et al., 2018). Therefore, it outperforms other deep neural network models (e.g., RNN, LSTM) in terms of efficiency and stability, providing a broader range of applications (He et al., 2019) (Li et al., 2017). This makes BERT particularly effective for tasks that require a deep understanding of complex relationships and entities (Lee et al., 2019). Researchers can fine-tune BERT using task-specific labeled data and apply it to specific domains (Vaswani et al., 2017) (Sun et al., 2023). Based on this, we designed an architecture to facilitate the automated construction of a fire-fighting spatial knowledge ontology, as depicted in Fig. 2. This architecture comprises three components: a task-specific dataset(a), triple extraction(b), and ontological storage(c). The primary focus of the research is on automating the acquisition of triples from text data. Subsequently, the automatically extracted triples are mapped to OWL files and converted to RDF format using the open-source toolkit RDF2RDF. The ontology is then stored by importing the RDF data into the graph database via Neosemantics.

The sliced single sentence is used as input to the ternary extraction module. We have introduced a novel entity-relation joint triple extraction model named Fire-GeoBERT, which is an advancement over the pre-trained BERT model, specifically optimized to better integrate the information between entities and their relationships. This model is fine-tuned for specific downstream tasks to automate ontology construction, replacing the traditional manual extraction method. It efficiently combines the information between fire equipment entities and their geographical spatial relationships, effectively handles the cross-nesting of entities, and accurately captures the spatial relationships between entities. The fundamental process is shown in Fig. 3(i e., a further particularization of Fig. 2(b). The extraction results are outputted in the format of two types of triplets as demonstrated in Table 3. In these triplets, “entity1” signifies the fire component entity, “relationship” denotes the spatial relationship, and “value” encapsulates the attribute value.

3.4.1. Fire entity recognition

The BERT pre-trained model segments and vectorizes the input sentence into a composite of word, sentence, and positional vectors. These are then processed via a bidirectional Transformer to extract features, creating a semantically rich sequence vector (Chomsky, 1957). Multi-head attention allows for parallel processing of the input sequence, detailed in formula (1).

$$multihead(Q,K,V) = concat(head_1,\ldots,head_n)W^O. \tag{1}$$

Where W^O is an additional weight matrix, as shown in formula (2) and formula (3).

$$head_n = attention\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{2}$$

$$attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)\cdot V \tag{3}$$

Among them, the matrices Q, K, V are the input word vector matrices of the encoder; d_k represents the number of columns in the $Q, K,$ and V matrices, which is the vector dimension. For an input sequence of length n, the vector corresponding to position t and ω_k are defined as shown in formula (4) and formula (5).

$$f(t)^{(i)} := \begin{cases} \sin(\omega_k\cdot t), i = 2k; \\ \cos(\omega_k\cdot t), i = 2k+1. \end{cases} \tag{4}$$

$$\omega_k = \frac{1}{10000^{\frac{2k}{d_k}}} \tag{5}$$

After the word sequence vector is subjected to max pooling, it is then concatenated with a special [CLS] vector to obtain the representation of the entity. The obtained entity representation x_e is shown in formula (6).

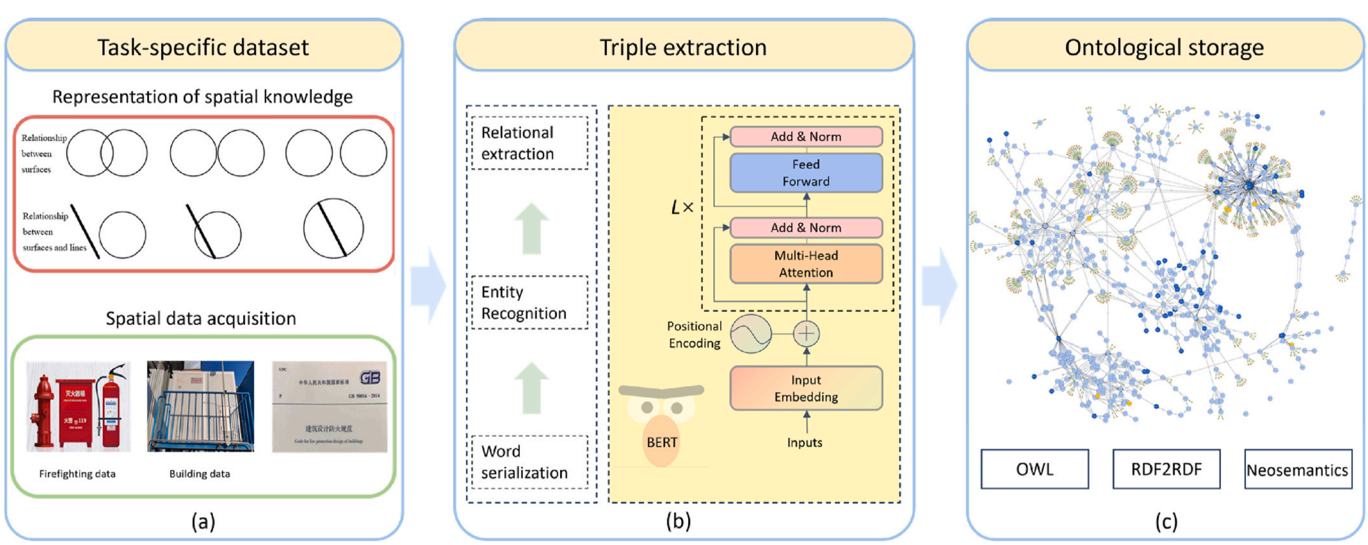

Fig. 2. General framework for ontology automation building.

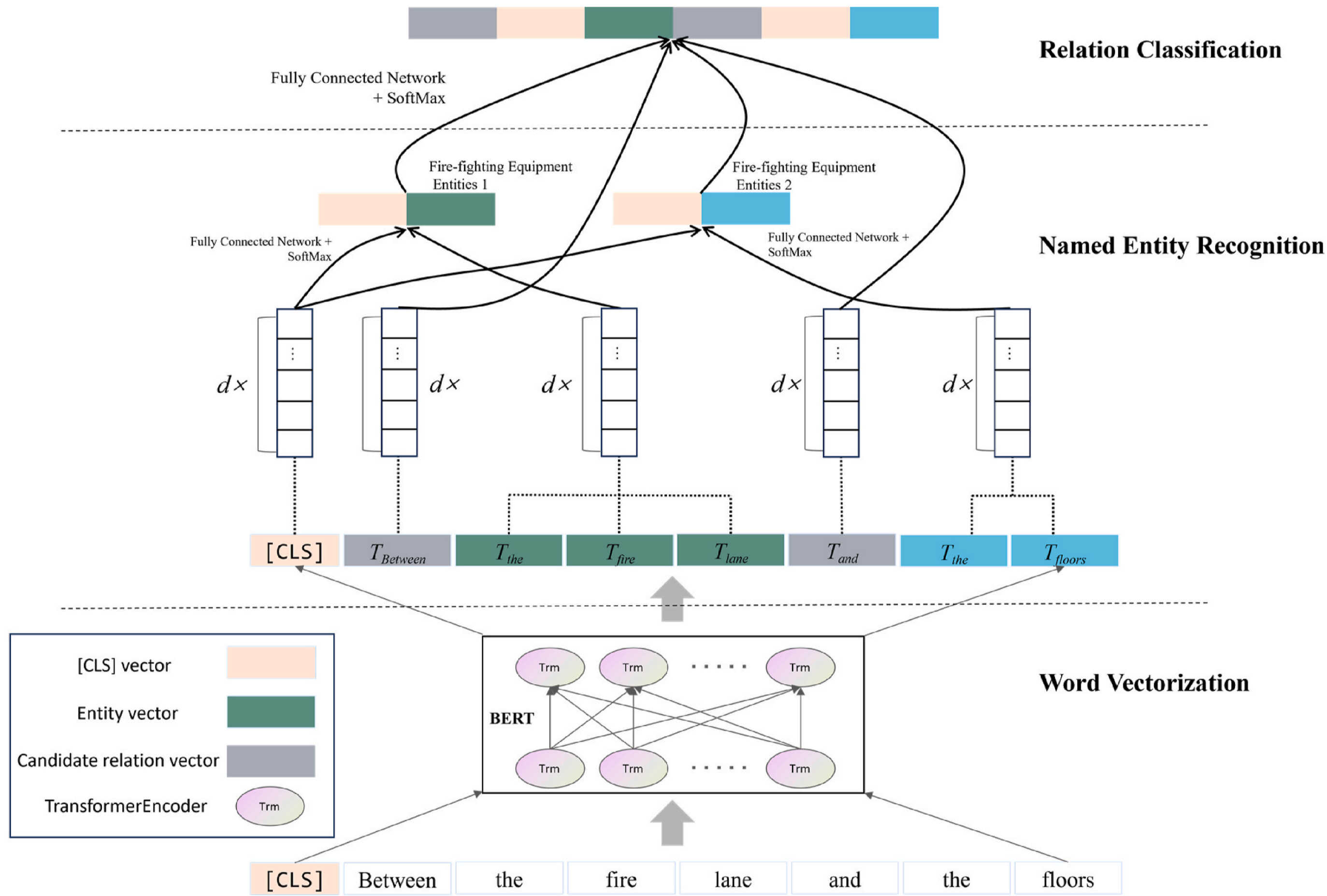

Fig. 3. Fire-GeosBERT model architecture.

Table 3
Two different representations of the fire space triad.

Format	Triples: [entity1, relationship, entity2]	Triples: [entity1, relationship, value1]
Instances	[Fire alarm display panel, RAbove, Floor]	[Fire partition, RSpatial, not be less than two]

$$x_e := \frac{\sum_{i=n}^{m} e_i}{m-n+1} \circ c \tag{6}$$

e_i represents the word vector representation after undergoing maximum pooling, while c represents the semantic representation of the entire text.

Finally, the entity vector x_e is passed through a fully connected network, added by a softmax activation function, to calculate scores for each category. This is illustrated in formula (7).

$$\hat{y}_e = softmax(Wx_e + b) \tag{7}$$

3.4.2. Spatial relation extraction

After extracting fire equipment entities, the next step involves extracting spatial relations between named entities using BERT. A set of predefined spatial relationship classes $SR = \{sr_1, sr_2, \ldots, sr_j\}$ is established. Spatial relationships are classified using BERT's [CLS] token and a sentence classifier with a single output layer. The input consists of anonymized target named entities represented by predefined labels (e.g., @objFP$, @objFE$) and the vector of the text between entities. This combined vector undergoes maximum pooling, is concatenated with neighboring entity vectors, and fed into a fully connected relationship classifier. This classifier computes class scores via a weighted sum, applying *SoftMax* to determine the relationship, as specified in formulas (8) and (9).

$$x_r = W_s(x_{e1} \circ c_r \circ x_{e2}) + b \tag{8}$$

$$\hat{y}_r = softmax(x_r) \tag{9}$$

Among them, c_r is the word sequence vector after the maximum pooling process, x_{e1} and x_{e2} are the vector representations of the fire entities before and after c_r.

3.4.3. Model performance evaluation

The Fire-GeosBERT model uses cross-entropy as its loss function to quantify the difference between predicted probability distribution and actual probability distribution. A lower cross-entropy value indicates better predictive performance of the model. The loss function for the RE task is given by formula (10).

$$L_r = \sum_{i=0}^{m} -r_i \log(\hat{y}_r) - (1 - r_i)\log(1 - \hat{y}_r) \tag{10}$$

m is the number of samples, and r_i is the one-hot vector mapped from the category label.

The loss function for entity relation joint extraction should be represented as the sum of entity recognition loss function and relation classification loss function, as shown in formula (11).

$$L = L_e + L_r \tag{11}$$

The precision (P), recall (R), and F1-score were used to evaluate each label. Finally, the weighted (micro) average F1-score was used as the standard to measure the overall performance of the model. The definitions of the evaluation metrics are as follows:

$$Precision = \frac{C_{TP}}{C_{TP+FP}} \tag{12}$$

$$Recall = \frac{C_{TP}}{C_{TP+FN}} \tag{13}$$

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{14}$$

$$weight\ F1 = \frac{\sum_i n_i F_{1,i}}{\sum_i n_i} \tag{15}$$

Table 4 displays the meaning of each parameter in formulas (12)-(15).

3.5. Modeling spatial relationship rules

While ontologies with triplets like < firewall, RInterlinked, exterior wall surface 0.4 m> and <firewall, RSeparation, corner > describe semantic relationships effectively, they alone do not fully depict the complex rules of fire regulations (Zhang and El-Gohary, 2021). Therefore, after extracting these triplets using the Fire-GeosBERT model, further interpretation into computer-readable formats such as logical clauses are necessary. This involves deeper analysis through spatial grammar parsing, formalization of spatial knowledge, and logic sentence generation, as illustrated in Fig. 4.

3.5.1. Space grammar parsing

Grammar parsing involves analyzing annotated sentences to generate syntax trees based on grammatical principles. Chomsky's theory of Universal Grammar categorizes grammar into four types: Regular, Context-free (CFG), Context-sensitive (CSG), and Unrestricted (Chomsky, 1957). CFGs, recognized for their robustness and simplicity, are extensively utilized for deriving tree structures from sentences and are specifically applied in this study to parse individual sentences.

Context-free grammar (CFG) is defined by the quadruple $G = \{N, \Sigma, R, S\}$, where N represents a finite set of non-terminal symbols, Σ a set of terminal symbols, R a set of production rules $R = \{r|r : N \rightarrow (N \cup \Sigma)^*\}$, and S the initial sentence symbol. Due to the specific needs of spatial rule analysis in firefighting, general Part-Of-Speech (POS) tagging is insufficient. This study introduces a specialized POS tagging adapted for firefighting spatial constraints, enhancing traditional POS tagging by focusing on spatial relationships among entities. Table 5, referenced from literature (Xu et al., 2019), in conjunction with the developed ontology model, introduces three supplementary methods for part-of-speech tagging representation.

The normative clauses are constrained according to spatial semantics. The attribute value constrained clauses and relationship constrained clauses are subjected to CFG syntactic analysis. The results are as follows:

(1) Attribute value constrain category.

This clause can be dissected into noun phrases and verb phrases based on morphemes. The noun phrases consist of several entities, corresponding to "linear beam smoke detector" and " Light beam axis " in the clause. The verb phrases can be further divided into MD, S, NP, and CPR. MD is a verb that includes deontic modality, corresponding to "should" in the clause. S denotes spatial information, which can be further delineated into a composition of a verb phrase (VP) and a noun phrase (NP), specifically VP + NP→S. CPR represents comparative relationship words, usually composed of prepositions (IN) and subordinating conjunctions (SCV), i.e., IN + SCV→CPR, corresponding to "within" and "the range of 0.3 m–1.0 m" in the clause. The corresponding context-free grammar representation is: $N = \{NP, VP\}; S = \{sentence\};$, as shown in formula (16) and Fig. 5.

$$R = \begin{cases} E \rightarrow Light\ beam\ axis; linear\ beam\ smoke\ detector; ceiling \\ S \rightarrow maintain; a\ vertical\ distance\ to \\ MD \rightarrow should \\ CPR \rightarrow within; the\ range\ of\ 0.3m\ to\ 1.0m \end{cases} \tag{16}$$

(2) Relationship constraints category

Relational constraint clauses are bifurcated into two types: those that delineate the relationship between entities using spatial verbs, and those that articulate the relationship using spatial prepositions. Concerning the former type, this clause is structured from noun phrases comprising two sets of morpheme information: entities and spatial verb constraints, which correspond to "fire hose reels", "high-rise civil buildings", and "installed in". However, an examination of these clauses suggests an additional implied meaning: the existence of a relationship wherein one fire component or architectural space area encompasses another fire component or space. Consequently, it becomes necessary to introduce the preposition terminal symbol IN to represent the implied relationship of inclusion. The SVN rule is constituted by combining the spatial verb SV with the preposition IN, resulting in SVN=SV + IN. The corresponding representation in context-free grammar is as shown in formula (17) and Fig. 6.

$$R = \begin{cases} E \rightarrow Fire\ hose\ reels; high - rise\ civil\ buildings \\ SVN \rightarrow installed; in \\ VP \rightarrow should; be \end{cases} \tag{17}$$

Regarding the latter type, this clause is composed of two noun phrases, specifically "firewall" and "floor", along with the directional word "above". The distinguishing feature of such clauses is the presence of directional words like "above", "below", "left", "right", which serve to specify the positioning of fire protection components. The corresponding representation in context-free grammar is as follows formula (18) and Fig. 7:

$$R = \begin{cases} E \rightarrow Firewalls; floor \\ SP \rightarrow above \end{cases} \tag{18}$$

3.5.2. Formalization of spatial knowledge

This section converts the results of CFG syntactic parsing into a hierarchical tree structure, extracting spatial cognitive linguistic elements such as distance values and comparative relations from various levels of the parse tree. Normative texts are arranged to reveal attributes and relationship constraints through elements like spatial prepositional phrases (CPR) "within the range of 0.3 m–1.0 m," verbs, verb phrases (SVN), or prepositions (SP) such as "installed," "installed in," or "above." These elements, tagged according to a self-defined function result list (e.g., CD, CPR, UN), represent spatial relationships between objects, formalized using an ontology outlined in section 3.4 that structures concept relationships in triples. This facilitates the development of a domain-specific tree structure model (E et al., 2019),

Table 4
Definition of the parameters of the evaluation indicators.

Parameters	Comment
C_{TP} (true positive)	The number of the accurate prediction of fire entities or spatial relationships.
C_{FP} (false positive)	The number of incorrectly predicted fire entities or spatial relationships.
C_{FN} (False Negative)	The number of fire entities or spatial relationships that exist in the corpus but have not been identified.

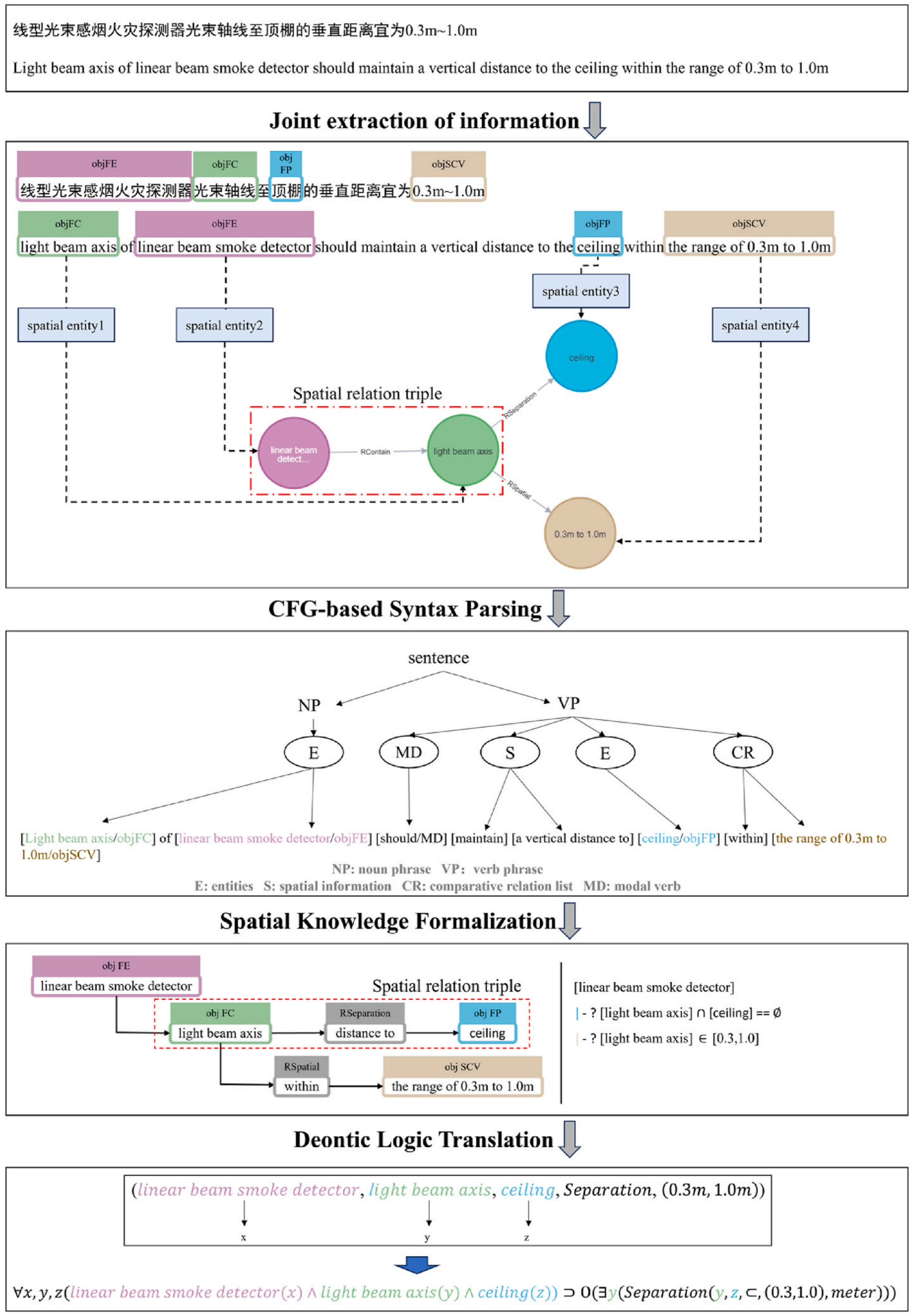

Fig. 4. The process of formalizing spatial knowledge into logical clauses.

Table 5
Newly added spatial vocabulary applicable to CFG.

Terminal Tokens	Tag	Example individuals
List of spatial verbs	SV	Bury/buried, install/installed, place/placed, locate/located …
List of prepositions of space	SP	Above, under, below, on, at, beyond, away from …
List of comparative relation	CPR	Less than, greater than, at least, at most, within a distance of …
List of spatial constraint value	SCV	the range of 0.3 m–1.0 m/obj SCV,greater than 2.2 m …

exemplified in Figs. 8 and 9 with two types of clauses, thus underpinning the fundamental tree structure model described.

In the attribute constraint clause, objFE acts as the parent class of objFC. The combination of [Light beam axis/objFC] and [ceiling/objFP], along with the spatial relationship 'RSeparation', forms a spatial relationship triplet, specifically {[Light beam axis/obj FC], R Separation, [ceiling/objFP]}. The spatial semantic element corresponding to 'RSeparation' is identified as 'distance to'. Consequently, this spatial semantic element is instantiated as the spatial relationship node 'RSeparation'.

In the clause on relationship constraints, there are directional words such as "above", "below", "left", and "right" that restrict the position of fire components. In the case of such clauses, given the absence of constraints on attribute values, the resultant tree structure does not incorporate the relationship denoted as "RSpatial". Nonetheless, the utilization of "left" and "right" presents considerable challenges, as these directional relationships fundamentally rely on a clearly defined coordinate system. In the absence of specified spatial coordinates, the interpretation of these terms becomes ambiguous, which limits their

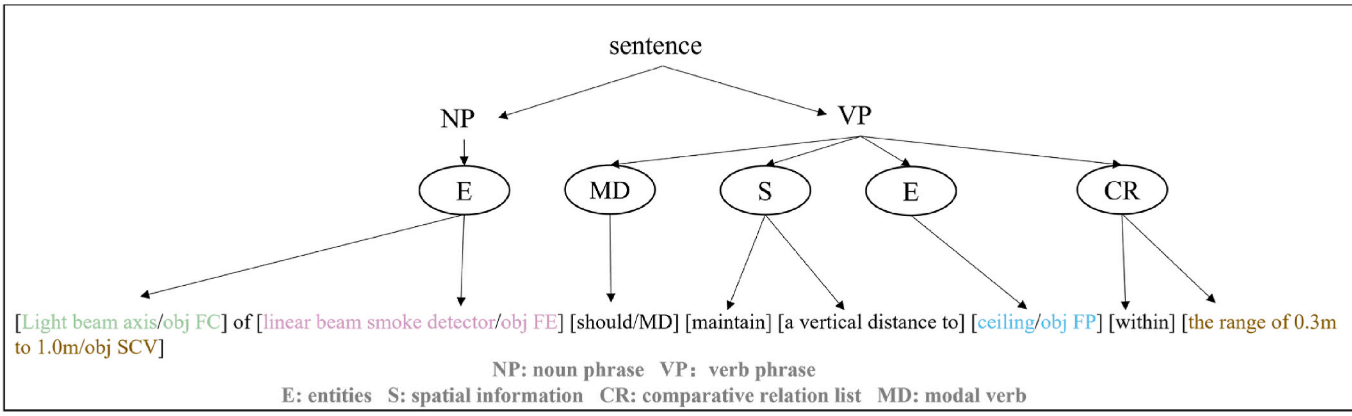

Fig. 5. Analysis results of attribute value constraint clause.

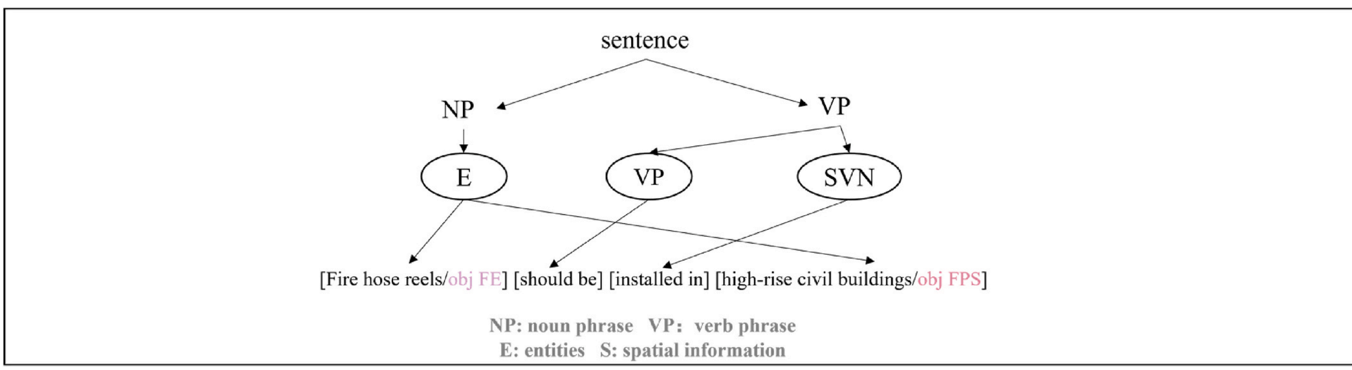

Fig. 6. Analysis of clause on inclusion relationships.

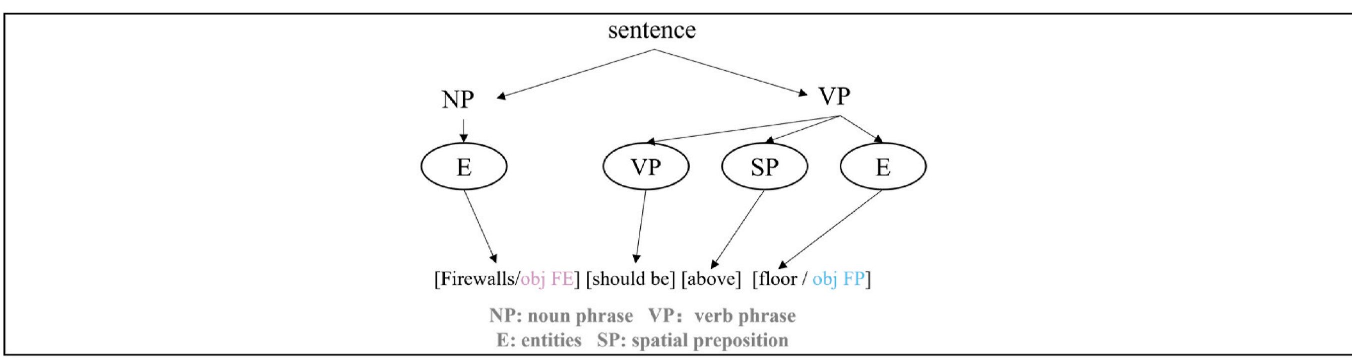

Fig. 7. Analysis of positional relationship constraint class regulations.

applicability within the framework of fire codes. This underscores the necessity for future research to establish robust spatial coordinate systems wherein directional relationships can be accurately evaluated through defined positive and negative values. This advancement will improve the clarity and functionality of orientation constraints for fire protection components in building information modeling.

3.5.3. *Logic sentence generation*

After extracting spatial cognitive linguistic elements and formalizing spatial knowledge from fire safety regulations, we propose using deontic logic language (DL) to structure semantic constraints. DL is an extension of first-order logic (FOL) that incorporates normative concepts, enabling the formal representation and specification of laws, legal rules, and precedents. It is specifically designed to articulate deontic reasoning and address ethical dilemmas by employing operators such as O (obligation), P (permission), and F (prohibition) to represent agents constrained by normative systems (Cheng, 2008). This framework enhances the effective articulation and computation of the logical implications of normative systems within regulatory contexts. Terms such as “must,” “should,” “may,” and “prohibited” are commonly found in fire code descriptions. These terms express deontic modality; for instance, “should” generally expresses obligation modality, while “may” denotes permission. Therefore, this approach is necessary to facilitate rule checking through a hierarchical grading system ranging from pass to fail and to introduce new methods of identifying, representing, and interpreting results.

The relationships between these deontic operators are detailed in formulas (19) and (20), enhancing the comparison of rationality and consistency across various deontic arguments.

$$F_{\varphi} := \neg P_{\varphi} \tag{19}$$

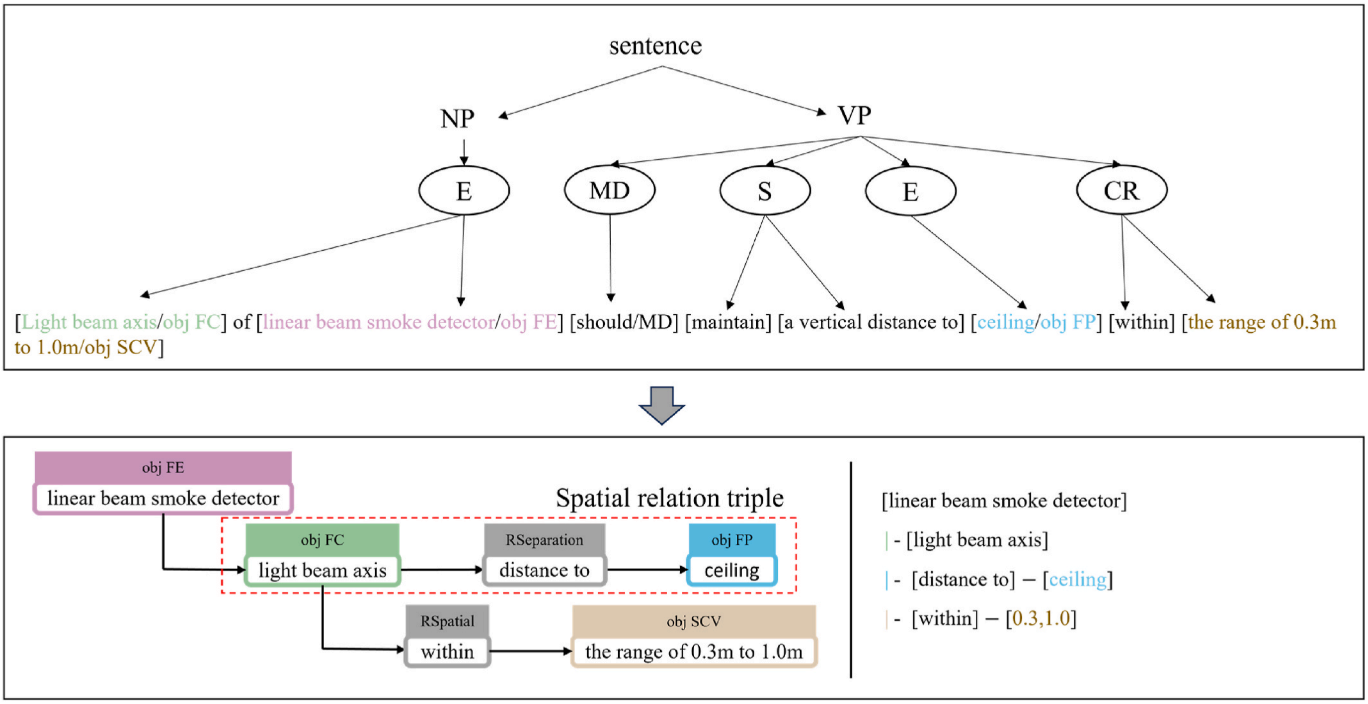

Fig. 8. The tree structure generated after formalizing the constraints of attribute values.

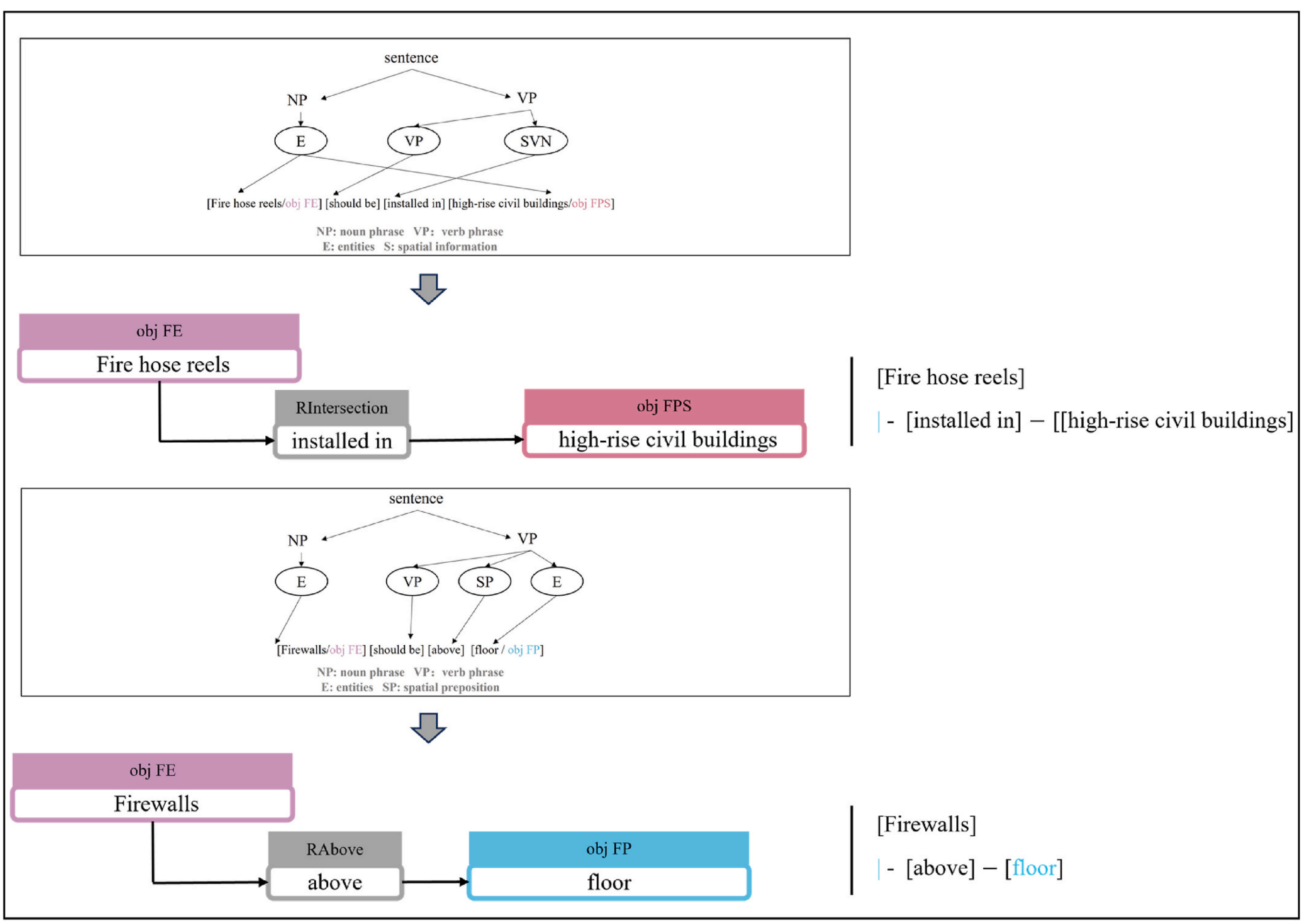

Fig. 9. The tree structure is generated after formalizing the constraints of spatial relationships.

$$O_{\varphi} := \neg P_{\neg\varphi} \tag{20}$$

Among them, formula (19) represents " φ is prohibited, i.e., φ is not allowed"; formula (20) represents " φ is obligatory, i.e., not doing φ is not allowed".

Based on Kim's extended DL representation method (Kim and Chi, 2019), we extract triplets in the form of logical clauses for automatic compliance rule checking in fire protection. DL statements comprise predicates and functions manipulated with deontic modal operators (obligation "O", permission "P", prohibition "F") and First-Order Logic (FOL) operators (conjunction "∧", disjunction "∨", negation "¬", implication "⊃"). Quantifiers ("∀" for universal and "∃" for existential) specify the scope and existence of variables within these statements.

- The deontic operator indicator aligns with the deontic modal operator, employed to define the normative mode of specific predicates or functions in Deontic Logic (DL) statements. For instance, "should be" correlates with the obligation operator "O". Thus, " $O\,(Separation\,(x,y))$" signifies that x and y are obliged to maintain a certain spatial relation as mandated.
- Regarding spatial relationships, including topological relationships and positional relationships, they can be expressed as predicates. For instance, the expression $Separation(y,z)$ in Fig. 4, where y,z represent two distinct entity variables. Based on this, the addition of spatial constraint relationships equates to $Separation(y,\ z,\ \subset,\ (0.3,\ 1.0))$, where "(0.3,1.0)" denotes the specified constraint value.

4. Experiments and results

4.1. Description

The compliance checking of fire design is critical for building fire safety and is enhanced by digital review discussions in documents from the State Council of China. These discussions support the "14th Five-Year Plan" for national construction industry development, pushing towards intelligent construction (Ni, 2021) (Xu et al., 2016). Our research focuses on fire safety, starting with creating a fire space dataset of 800 normative phrases and their semantic annotations from regulations like the "Building Regulations." (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2020) We analyzed and evaluated the constructed ontology knowledge base, formalized the spatial knowledge, assessed its correctness, and then translated it into deontic logic clauses. These were applied in Revit BIM models to verify the practical applicability of our method.

4.2. Semantic labeling scheme

We propose a semantic labeling scheme tailored for relation extraction, which assigns semantic labels to candidate entities and relations in a corpus based on their roles within the defined semantic structure. Our predefined schema collection categorizes relations P, and their corresponding subjects S and objects O. Relations are divided into two types based on the complexity of O values:

(1) simple O values, exemplified by the "RAbove" relation defined as $\{S\_TYPE{:}objFE, P: RAbove, O\_TYPE: \{@value: floor\}\}$, where O is a single text fragment.
(2) complex O values, exemplified by the "RSpatial" relation defined as $\{S\_TYPE: objFC, P: RSpatial, O\_TYPE: \{@value: 0.3m \sim 1.0m, inWork: Ceiling\}\}$, where O comprises multiple semantically explicit text fragments filling multiple slots in the structure. The "@value" slot is considered the default and indispensable O value slot, while other slots are optional, as illustrated in Table 6 with examples of our schema.

Table 6
Schema examples in our dataset.

S_TYPE	Predicate	O_TYPE	SPO example
objFE (Fire equipment)	RAbove	objFP (Floor plate)	S:火灾报警控制器, P:高出, O:地(楼)面 S: fire Alarm Controller, P: higher, O: floor
objFC (Fire component)	RInterlinked	objFP (Floor plate)	S:探测器, P:安装, O:走道顶棚 S: detector, P: installed, O: ceiling of the aisle
objFC (Fire component)	RSeparation	objWall (Wall)	S:探测器, P:距离, O:端墙 S: detector, P: distance, O: boundary wall

4.3. Dataset development

The dataset comprises of **800 sentences (3160 elements)**, which are categorized into two distinct types: attribute value constraints and relationship constraints. These sentences are extracted from the "Building Specifications" source. Moreover, the dataset encompasses 8 types of entities and 6 types of spatial relationships. To establish a reliable benchmark for evaluating the entity relationship joint extraction model, the 800 individual sentences within the dataset were meticulously annotated by multiple annotators. During the annotation of the test dataset, meticulous attention was paid to ensuring annotation accuracy by developing detailed and specific criteria that clarified the definitions and application scopes of each semantic label. Before formal annotation, two rounds of trial annotations were conducted to refine the guidelines and minimize ambiguities arising from varying expert interpretations of natural language (Clark and Creswell, 2008). Each instance was initially assigned to two annotators with general engineering experience. A third annotator, a professional from the relevant fire service, subsequently reviewed the instances with consistent answers to ensure data compliance and logical coherence. This intentional decision aimed to incorporate perspectives from individuals less familiar with the concept of firefighting literacy. This approach was designed to simulate potential interactions that less technically experienced users might have with firefighting knowledge tailored for small, specialized populations (Zhang and El-Gohary, 2023) (Etikan, 2016). Subsequently, to ensure a balanced distribution of various semantic entities, data statistics were regularly monitored and analyzed throughout the dataset development phase. Finally, all instances deemed correct were compiled to form the final dataset. During the validation phase of the gold standard, annotators assessed annotation correctness based on three specific criteria: (1) Clues should be derived solely from the provided sentences, irrespective of the real-world veracity of the triples; (2) Subjects and objects must align with the intended types defined in the architecture; (3) Predicates need not be explicitly mentioned in the sentence. Each expert completed the annotation independently, and the initial inter-annotator agreement, measured by the F1 score, was **82%**. This indicates that the manual annotation process for entity-relationship pairs demonstrates strong consistency, reliability, and repeatability. Therefore, the quality of manual annotation in preparing the test dataset was considered high (Etikan, 2016) (Pestian et al., 2012). Fig. 10 presents an example of an annotation candidate, representing entities and relationships within the statutory text.

4.4. The result of automated construction of ontology

4.4.1. Model training and testing

We conducted experiments using a specially constructed dataset, divided into training, validation, and test sets in a 6:2:2 ratio, to verify the model's effectiveness. Before training, we set several key hyperparameters based on a minimum tuning strategy recommended by related studies (Eberts and Ulges, 2020) (Ferrell et al., 2022). We experimented with four training batch sizes and two evaluation batch

根据语句：消防水泵房应至少有一个可以搬运最大设备的门

According to the sentence "Fire pump rooms should have at least one door capable of transporting the largest equipment."

是否可以判断：<消防水泵房>(objFPS)应<至少有一个>(RContain)搬运最大设备的<门>(objDoor)

Judge whether it is correct: <Fire pump rooms>(objFPS) should <have>(RContain) at least one <door>(objDoor)

Fig. 10. An example of annotated dataset.

sizes. For the fine-tuning task, batch sizes of 4, 8, and 16 were found to be appropriate. The number of training epochs was limited to 60 to ensure convergence on 12 GB GPUs within this timeframe (He et al., 2019). Additionally, we employed the AdamW optimizer for gradient computation and parameter updates. To reduce overfitting and enhance stability, we tested initial learning rates of 1e-5, 2e-5, 3e-5, and 4e-5, and implemented a learning rate optimization strategy that included a linear warm-up phase. The results indicate that a learning rate of 2e-5 and a batch size of 4 yield the best performance in our study, with the model achieving convergence within 60 epochs and the loss value reduced to a satisfactorily low level. This methodical approach, incorporating both initial recommendations and customized tuning based on model performance, ensured a robust training process. The testing process mirrored the training procedure, with the sole exception that the gradient was set to zero during backpropagation.

4.4.2. The results of joint extraction of entities and relationships in fire regulatory documents

The performance of the entity-relation joint extraction model on the validation set is shown in Table 7, Table 8 and Table 9. Firstly, Table 7 is analyzed, presenting the classification outcomes for individual Named Entity Recognition (NER) tasks. The training process involved a total of 1945 entities, and the weighted (macro) average F_1 score for this task is **86.27**, with a recall rate of **84.64**. This is a very promising result as it demonstrates the ability to identify different semantic labels in fire safety regulations with relatively high accuracy. Tables 8 and 9 compare the results of relation extraction. Table 8 presents the results of a single RE task, without considering the correlation between individual relations and their corresponding candidate entity pairs. This approach represents a more traditional method of relation extraction. The RE task for the 6 types of relations achieved a weighted (macro) average F_1 score of **79.25**. Conversely, Table 9 represents the results of a relation classifier that extracts candidate entity pairs and determines their association with the corresponding relation R. In comparison to Table 8, Table 9 attains a weighted (macro) average F_1 score of **81.81**, marking an improvement of **2.56**. This result demonstrates the feasibility of joint entity extraction and relation extraction.

Fig. 11(a) illustrates the variation in training loss across epochs, with the model achieving convergence at epoch 30. The results of the spatial relationship classification are displayed in the confusion matrix (Fig. 11 (b)). The small squares on the diagonal represent the percentage of true positive samples to the total number of samples. The small squares on the diagonal line are notably darker, indicating the high effectiveness of our model. The darkest squares, located at the intersection of the first row and the first column, suggest that sentences labeled "R space" typically exhibit more prominent text features. However, the model still incorrectly classifies "RAbove" and "RContain" as "RSpatial." This misclassification may stem from the fact that "RSpatial" labels often include specific numerical constraints, indicating an area where the model's performance could be further enhanced. Furthermore, the limited size of our dataset—consisting of only 800 sentences—restricts the model's exposure to diverse examples, thereby impairing its ability to generalize effectively. This combination of complexity and limited training data underscores the challenges in achieving higher accuracy and emphasizes the necessity for further research to enhance model performance in future studies.

Table 7
Firefighting entity recognition results score statistics.

label	precision	recall	F_1	number of instances
objWall	87.00	88.78	87.88	35
objDoor	86.96	86.96	86.96	19
objFE	87.60	83.55	85.53	95
objFC	89.37	82.53	85.81	182
objFA	93.55	87.88	90.62	15
objSCV	83.81	85.15	84.47	92
objFP	91.07	77.86	83.95	45
objFPS	85.39	84.44	84.92	68
micro	**87.29**	**83.70**	**85.46**	**551**
macro	**88.09**	**84.64**	**86.27**	**551**

Table 8
Statistics on scores of spatial relationship classification results (entity type is not considered).

Label	Precision	Recall	F_1	number of instances
RSpatial	72.95	82.95	77.63	88
RIntersection	78.31	77.38	77.84	23
RInterlinked	86.47	80.33	83.29	36
RSeparation	92.16	69.63	79.32	42
RContain	81.06	83.81	82.41	87
RAbove	81.08	69.77	75.0	16
micro	**79.64**	**80.49**	**80.07**	**292**
macro	**82.01**	**77.31**	**79.25**	**292**

Table 9
Statistics on scores of spatial relationship classification results (entity type is considered).

Label	Precision	Recall	F_1	number of instances
RSpatial	79.16	82.43	80.76	88
RIntersection	79.27	77.38	78.31	23
RInterlinked	91.30	80.33	85.47	36
RSeparation	94.00	69.63	80.00	42
RContain	91.19	83.81	87.35	87
RAbove	90.91	69.77	78.95	16
micro	**86.30**	**80.33**	**83.21**	**292**
macro	**87.64**	**77.22**	**81.81**	**292**

4.4.3. Comparison with the state-of-the-art

To thoroughly evaluate our proposed Fire-GeosBERT model, we conducted comparisons with several leading entity and relation extraction models, including RNN (Sherstinsky, 2020), BERT-BiLSTM (Song et al., 2020), BiLSTM + SDP (Li et al., 2017), and Multi-head (Bekoulis et al., 2018), which is currently state-of-the-art model for joint entity-relationship extraction. The models mentioned above were evaluated individually using the datasets we developed.

Table 10 shows the test set evaluation results for our datasets, and Fig. 12 illustrates the comparison of the precision-recall curves for the five models. As expected, Fire-GeosBERT consistently surpasses existing state-of-the-art models in entity and relation extraction. Its Average

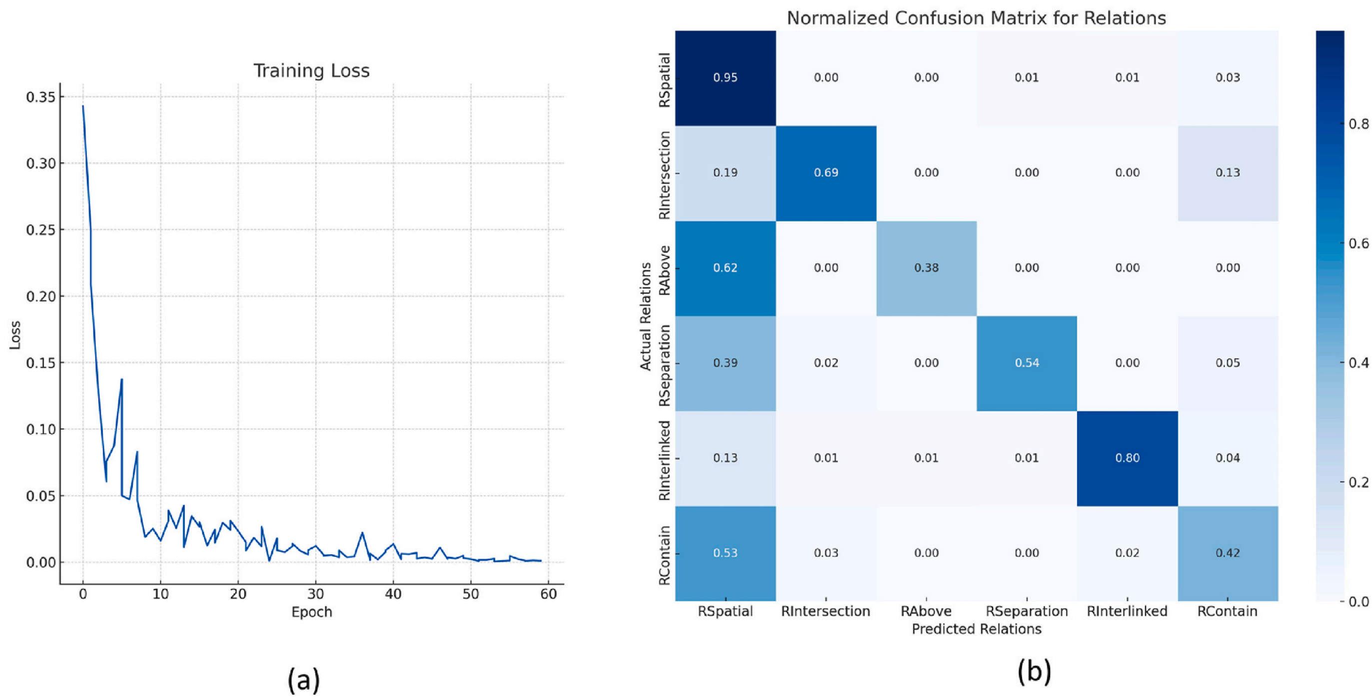

Fig. 11. The variation of training loss and confusion matrix of multi-class relation classification model.

Table 10
Test set results on our datasets, Fire-GeosBERT model achieves the state-of-the-art on both entity and relationship joint extraction tasks.

Model	Entity			Relation		
	Precision	Recall	F1-score	Precision	Recall	F1-score
RNN	64.89	74.24	69.27	43.95	67.44	53.22
BERT-BiLSTM	71.48	82.85	76.75	49.81	66.34	56.90
BiLSTM + SDP	78.27	85.50	81.65	65.80	74.35	69.82
Multi-head	85.42	83.07	84.14	76.74	77.27	76.51
Fire-GeosBERT (ours)	**88.09**	**84.64**	**86.27**	**87.64**	**77.22**	**81.81**

Precision (AP) values, shown in Fig. 12, confirm its superior performance over other deep learning models. The entity recognition performance has improved across all datasets; for instance, F_1-scores increased by **2.13%** on our dataset. Relational extraction shows even greater performance gains, significantly outperforming models such as RNN, BERT-BiLSTM, and Multi-head. This superior performance can be attributed to Fire-GeoBERT's integration of BERT's robust contextual embeddings with geometric deep learning techniques, effectively capturing the complex dependencies and spatial relationships between entities. The model not only utilizes the rich semantic context provided by BERT but also enhances it with a spatially-aware layer, allowing for a more nuanced understanding and representation of spatial relationships that are critical in domains such as fire safety regulations. In our dataset, Fire-GeoBERT significantly outperforms the current best model in correlation extraction scores, demonstrating its strong generalizability across various contexts and datasets—a crucial advantage for real-world applications where variability in textual structure and semantic richness greatly influences entity and relationship extraction outcomes.

Fig. 13 shows the result of the automated ontology construction. Following the construction of the main part of the ontology, it is essential to verify its consistency by ensuring that both the syntax and semantics of the domain ontology meet specified requirements. We employed the HermiT reasoner, included with the Protégé software, to conduct this verification. The reasoning process revealed no semantic errors. Subsequently, three domain experts with extensive ontology-related research experience were invited to manually construct the ontology based on the same content. Therefore, the domain ontology generated by domain experts can be regarded as the gold standard. Comparing our automated constructed ontology with the ontology manually constructed by the experts, instances and attributes reached 82.3% and 84.5% accuracy in automatic extraction, respectively. Finally, following the procedures outlined in Fig. 2(c), the ontology is stored in a Neo4j graph database. In this database, nodes of various colors represent different types of entities, and these entities are interconnected by directed line segments that indicate relationships. With this step, the automated construction of the ontology was completed.

4.5. *The results of grammar parsing*

The process of grammar parsing is implemented using the Python language and calls upon the Natural Language Toolkit (NLTK), from which the Context-Free Grammar (CFG) module is imported. During the experiment, two categories of fire space triplets are extracted from the automated ontology. These triplets, endowed with semantic labels, serve as the input. A tree structure model is constructed based on the containment relationship among entities, succeeded by the translation into deontic logic, culminating in the generation of corresponding deontic logic clauses. Two aspects of analysis and evaluation were performed: (1) whether the semantic expressed by the tree structure constructed based on the triplets is consistent with the original sentence, and (2) whether each element in the tree structure is correctly mapped to the deontic logic clause. The experiment was conducted on a 5.6 GHz Inter Core i9 CPU.

Fig. 14 shows the generated results of the syntax tree structure models for four sentences, including two simple structures and two complex structures. The sentences were translated from Chinese to English, maintaining the original word order. The first two sentences are complex sentences, while the latter two are simple sentences. For the complex sentences, the containment relationships between all objFE and objFC are correctly parsed, indicating that the obtained results can effectively express objFE as the parent node of objFC, with the parent and child nodes connected by the '|'. For the two simple sentences, the

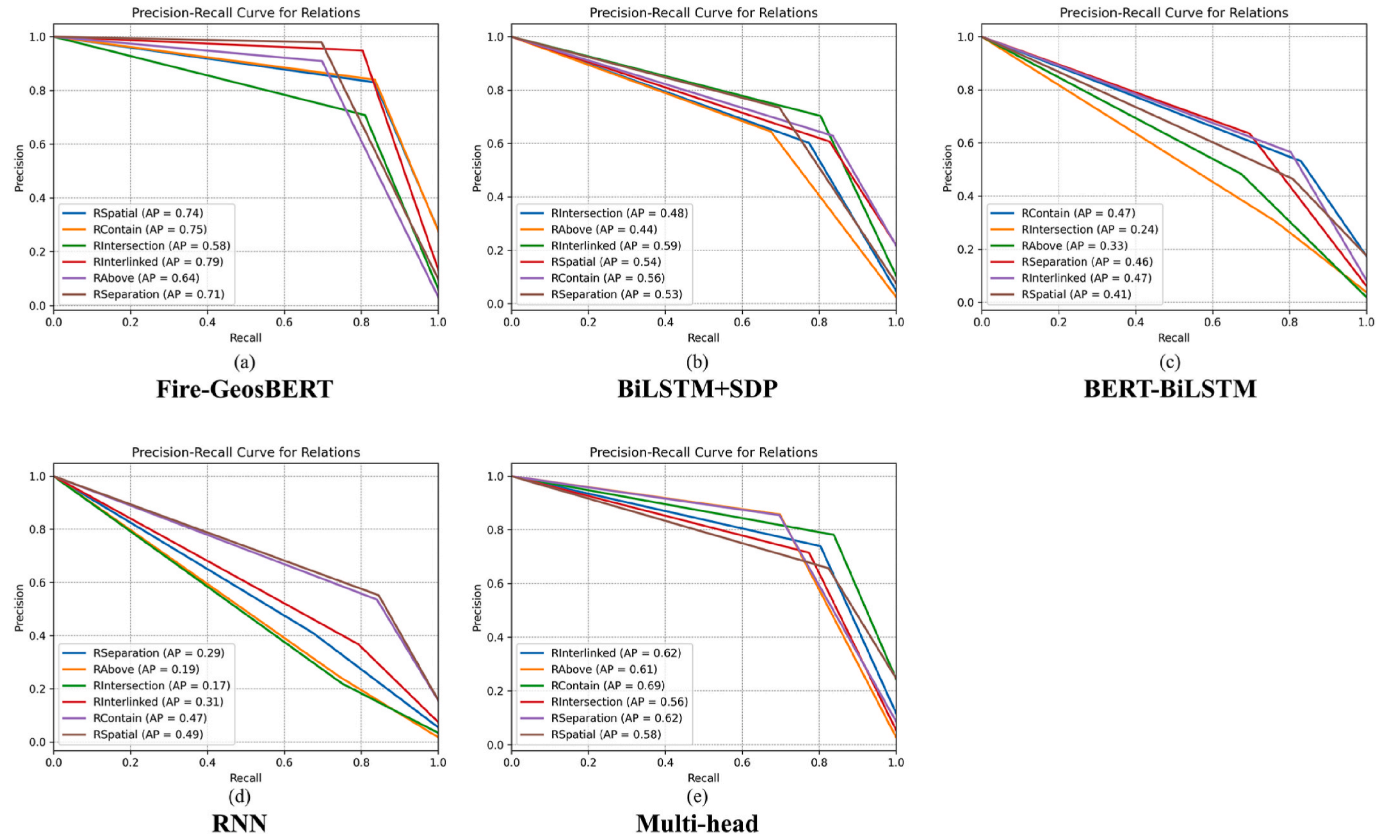

Fig. 12. Comparison of precision-recall curves of different models on our dataset.

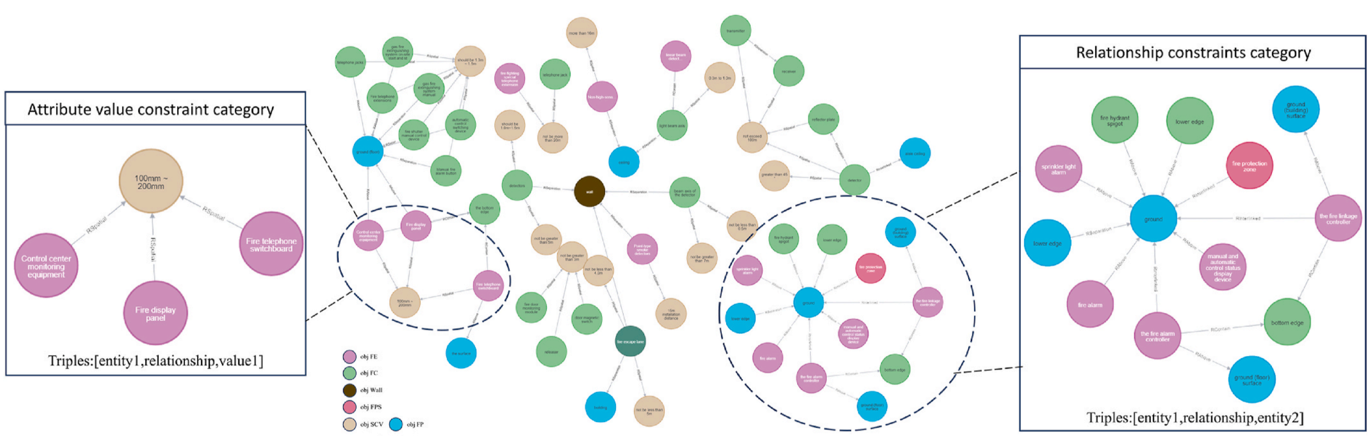

Fig. 13. Consolidated fire space knowledge ontology.

head entity is established as the root node, with all related triples identified and appended as child nodes. All four sentences shown can be correctly interpreted, but the spatial verbs representing relationships are 'null'. Consequently, the formalization of spatial knowledge utilizing Context-Free Grammar (CFG) is executed, accompanied by the semantic supplementation of absent verbs representing spatial relations. A syntax tree, predicated on CFG, is constructed in accordance with the definition of spatial relation verbs. Subsequently, the spatial relation semantics of the triplets are populated based on this syntax tree. The results show that the relationship part of the triplets' output by the four single sentences has been filled with corresponding verbs.

Following the formalization of spatial knowledge, information related to normative modalities and hierarchical structures within spatial constraints is methodically formalized through deontic logic. Ultimately, it is transformed into deontic logical clauses. Fig. 15 provides an illustrative example of how ethical logical clauses are formulated based on formalized spatial knowledge. Table 11 presents a comparison between our decoding methods and the state-of-the-art at the elemental decoding level. Zhang et al. proposed a rule transformation method utilizing regular expression-based pattern matching, while Xu et al. developed a rule formalization method grounded in logical representations. It is important to note that these methods employ encoded pattern matching rules that may not be directly applicable to other contexts, potentially limiting their generalizability. The results indicate that our method, achieving an accuracy of **96.26%**, surpasses existing state-of-the-art methods. This superior performance attests to its high automation and broad applicability, primarily due to: (1) the development of an ontology domain knowledge model that

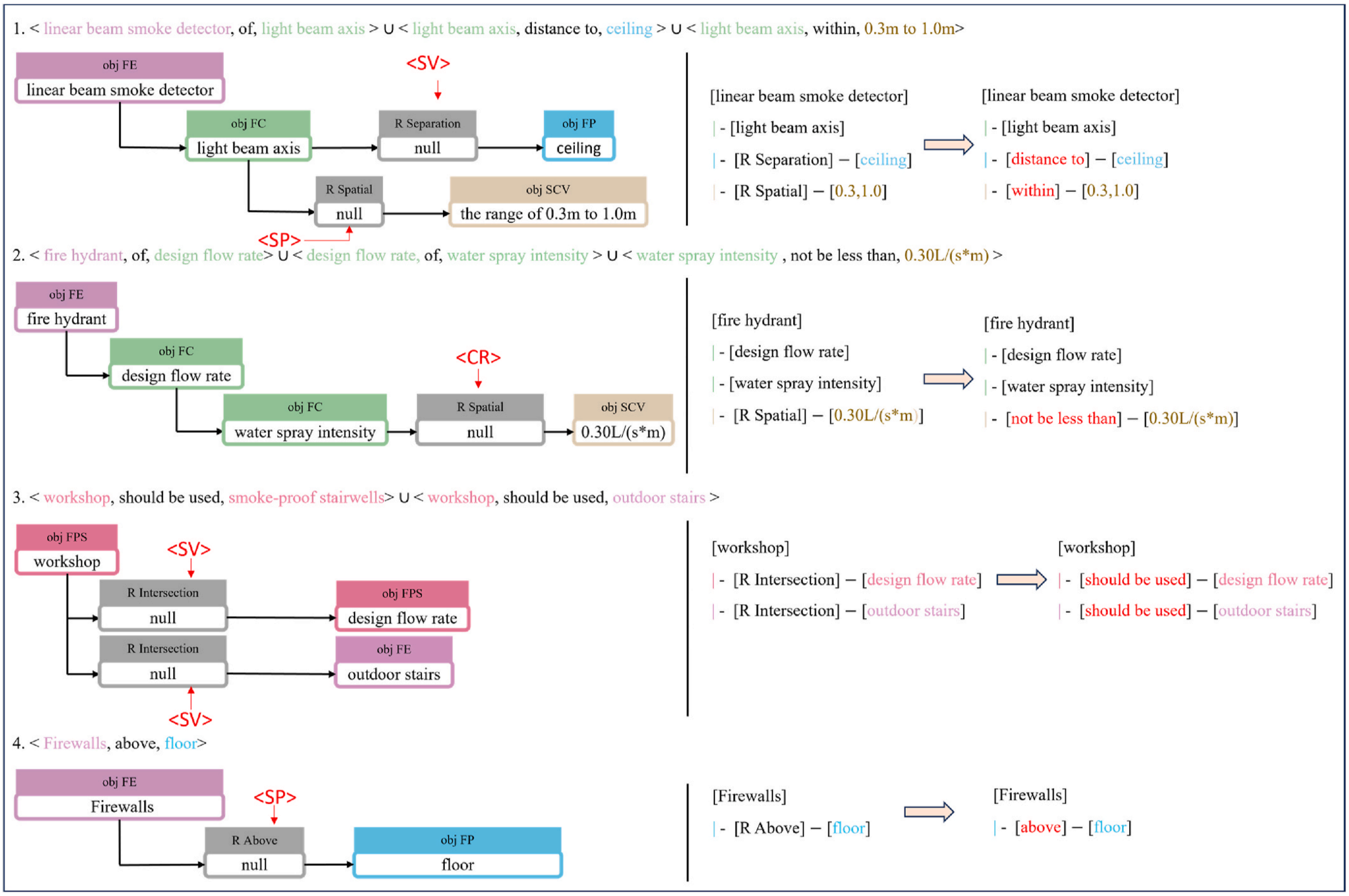

Fig. 14. The process of completing the relationship predicate based on CFG to generate tree structure.

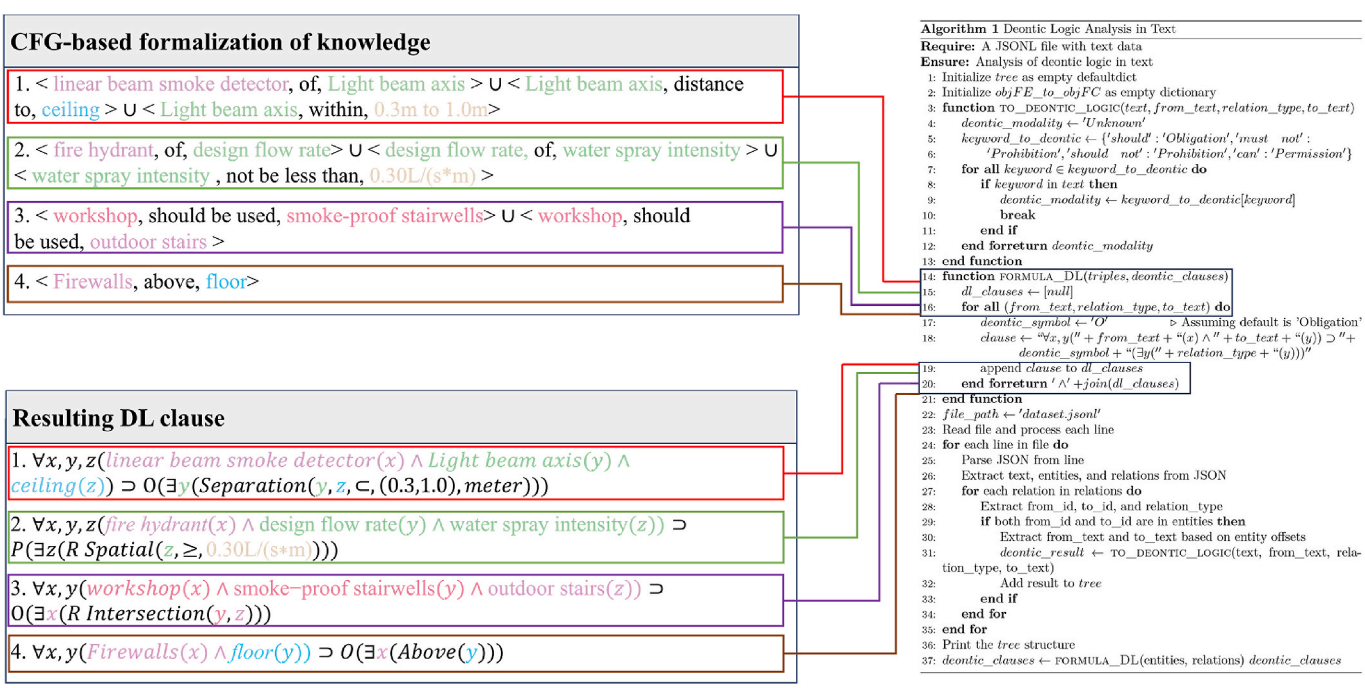

Fig. 15. The transformation process of formal spatial knowledge into logical clauses.

Table 11
Performance of logical clause conversion.

Method	Total element number	Correctly formalized elements	Accuracy
(Xu and Cai, 2021)	395	373	94.43%
(Zhang and El-Gohary, 2015)	1083	1030	95.11%
ours	**3160**	**3042**	**96.26%**

explicitly defines domain-specific concepts and their relationships, including implicit spatial relationships. (2) We use a larger dataset than those used in most existing studies. (3) Using fewer labels to represent different elements in a sentence greatly reduces specificity.

4.6. The results of rule checking

To verify the effectiveness of the proposed method, we applied the generated deontic logic clauses to the actual BIM models in Revit for rule checking. Revit Model Checker (Autodeska) is a plugin tool that can automatically check the compliance of BIM model elements and generate accuracy reports. We referenced the approach from the literature (Zhou et al., 2022) and converted the deontic logic clauses into XML format that can be recognized by Revit Model Checker. This XML format includes mappings of all filtering conditions of the deontic logical clauses, which are integrated into the Checkset. Readers are encouraged to refer to additional sources for comprehensive information on XML Checkset (Autodeskb).

Diverging from existing literature (Zhou et al., 2022), the XML Checkset developed in this study encompasses a broader range of highly specific filtering conditions, thoroughly applying deontic modal logic to the process of rule checking, thereby extending its applicability. Figs. 16 and 17 shows a general example of mapping deontic logic clauses to the XML Checker set, mapping relational constraint clauses and attribute value constraint clauses to the specified attribute names in the XML Checker. For instance, in the case of relational constraint clauses, the "Firewalls" element corresponds to "OST_Walls", the "floor" element to "OST_Floors", the spatial relationship "above" that constrains these two elements to "GreaterThan", and the deontic modal logic verb "obligation" is equated to a "True" value. Conditions leading to the failure of the check are categorized as "FailNoElements", implying that failure ensues if no corresponding elements are identified. Regarding attribute value constraint clauses, specific numerical constraints necessitate the utilization of designated parameter names and values. For example, "from 0.3 m to 1.0 m" has two numerical constraints, namely greater than 0.3 m and less than 1.0 m, which can be represented by "GreaterThan" and "LessThan" respectively. In addition, Revit defaults to recognizing imperial units when processing XML Checksets, so metric unit attribute values need to be converted to imperial units, for example: 0.3 m is approximately equal to 0.984252 feet.

Fig. 18 illustrates the outcomes of integrating the XML Checkset into the Revit Model Checker plugin for executing rule checks within the BIM model. As depicted, the review plugin is capable of directly opening the XML file that contains domain-specific rules and displaying the detailed texts of these rules. We selected five example rules to review a four-story apartment BIM model. The results indicated failures in the first and third inspections, yielding an overall accuracy rate of 60%. The results indicate that using the Revit Model Checker plugin for rule checks can identify components that do not meet the requirements and improve the efficiency of fire review.

5. Conclusion, limitations and future work

5.1. Conclusion

5.1.1. Contributions to the body of knowledge

In this study, we have developed a high-performance spatial semantic constraint rule checking methodology specifically tailored for the fire protection domain, representing a notable improvement in compliance verification methods. Our innovative model incorporates a dual fire protection regulatory framework and illustrates a spatial

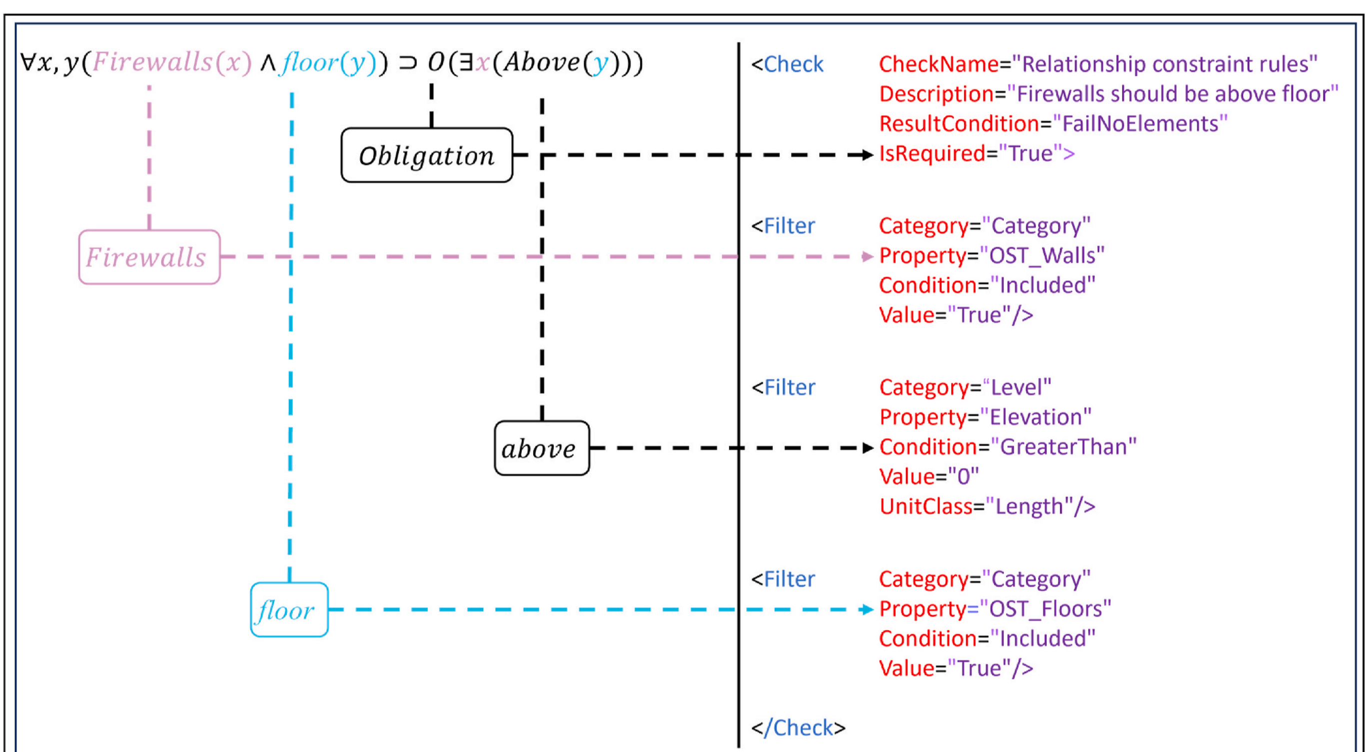

Fig. 16. Mapping relational constraint logic clauses to XML Checker set.

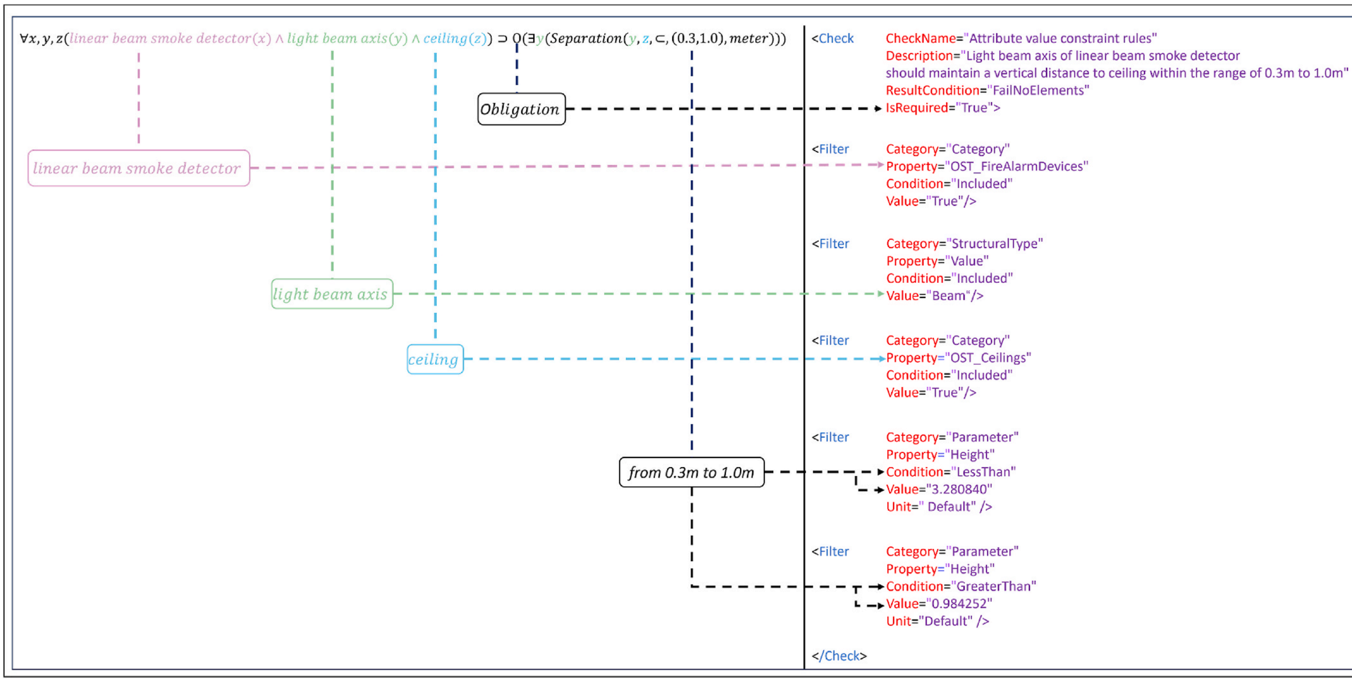

Fig. 17. Mapping attribute value constraint logical clauses to the XML Checker set.

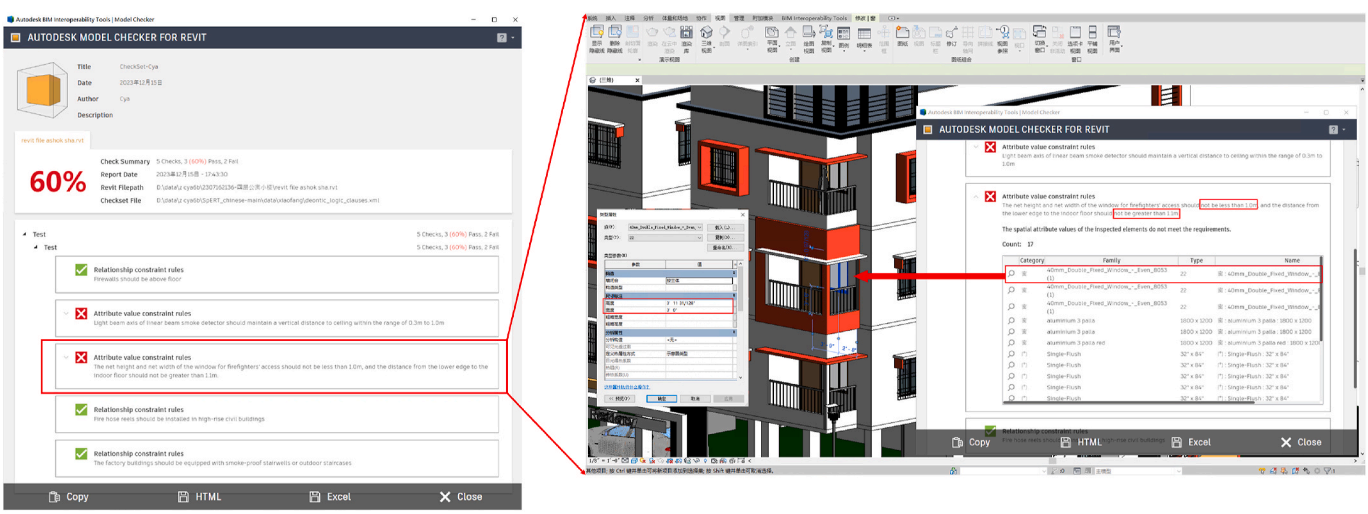

Fig. 18. Visualization of rule checking results.

knowledge representation crucial to the fire safety domain. The key advancements of our approach, in comparison to the state-of-the-art, are outlined below:

(1) We have developed a new fire code dataset that includes a comprehensive set of semantic and spatial labels. It encompasses eight fire-type entities and six medium spatial-type relations, facilitating the automatic extraction of spatial relation triples. Building on this, we propose a novel model for the joint extraction of information, streamlining the creation of domain-specific ontologies and significantly improving the automated compliance checking process. This approach not only accurately categorizes fire protection entities and their spatial interactions but also streamlines the identification and extraction processes through the joint model. The model's versatility is evident in its improved performance in recognizing relationships once candidate entities are integrated, achieving an F1-score of **86.27%** for entity recognition and **81.81%** for relationship extraction. This highlights its effectiveness in mapping spatial interrelationships within the ontology. Utilizing ontology-based representation for spatial knowledge structures standardizes specification drafting and greatly enhances the reliability and consistency of compliance checks.

(2) We introduce a high-performance approach to knowledge formalization. Context-Free Grammar (CFG) is employed to parse individual sentences and extract spatial semantic information. This information is then mapped onto the corresponding triples through an ontology. Our study explores the formalization of spatial knowledge, parsing sentences represented by triples into a tree structure. Furthermore, for each sentence containing modal verbs, we transform the formalized spatial knowledge into corresponding deontic logical clauses to achieve semantic and logical formalization. Our approach achieves high accuracy (**96.26%**) in **3160** spatial knowledge element formalization

tasks, all of which outperform existing knowledge formalization methods and are well-suited for generating rules related to the spatial configuration of firefighting components.

5.1.2. Practical implications

Overall, the ontology-based tree structure effectively formalizes spatial knowledge and standardizes the drafting of specifications. The resulting description logic clauses objectively and uniformly emphasize the importance of various provisions in the fire inspection process. This approach not only significantly contributes to the long-term stability of fire protection equipment and enhances fire safety management but also boosts public understanding of fire safety regulations. Looking ahead, based on the findings of this study, designers can apply our methodology to parse the structure of other fire protection prescriptive documents, provided these documents articulate requirements or rules similarly (i.e., through an object hierarchy and quantitative spatial requirements applicable to specific objects). Consequently, this methodology can guide the drafting of fire regulation texts and facilitate the semi-automatic parsing of irregular sentences. By demystifying complex regulations, it empowers a broader range of stakeholders, including non-specialists, to actively engage in discussions and decision-making processes related to fire safety management (Ma et al., 2022b). This inclusive approach promotes the continuous enhancement of fire safety regulations, thereby establishing a robust foundation for the intelligent management of the built safety environment.

5.2. Limitations and future work

This is an exploratory study aimed at validating the conjecture that ontology- and natural language-based learning can perform initial rule-based formalized decoding of unstructured fire codes. Therefore, the results obtained are solely focused on demonstrating the decoding of fire codes in China as outlined in this paper. It cannot be asserted that the proposed method can address the requirements of various types of code documents in all countries. Additionally, our research has identified some limitations that require enhancement in future studies.

There are still some aspects of this study that need improvement. Firstly, the study identifies a considerable challenge in how regulatory requirements are depicted in regulations, primarily through diagrams and tables. There is a critical need for developing computational methods that enable machines to interpret these graphical and tabular representations directly. This could involve advanced image recognition and processing techniques that convert visual data into machine-readable formats, eliminating the reliance on human translation into text. Developing algorithms capable of understanding and categorizing visual information such as diagrams and tables will be crucial. In addition, performance-based rules pose additional challenges in terms of data complexity and require a more nuanced approach to integrate into the system. Secondly, expanding the ontology and dataset is essential for broadening the framework's scope to include regulatory requirements across various sectors, including energy, electromechanical systems, and manufacturing. Currently, our dataset contains only 800 sentences, and the scalability of our rule-based NLP approach is inherently limited by the need to develop additional rules as the dataset grows. Enhancing the ontology requires the integration of additional domain-specific knowledge and relationships, which will enable the model to comprehend and process a broader range of regulatory texts. One of our future directions is to explore alternative methods to overcome this scalability limitation, while also expanding the dataset to improve the model's performance and accuracy. Thirdly, delving into more intricate syntax tree structures is crucial to accurately represent the semantic relationships in complex fire safety specification statements. Fourthly, utilizing pre-trained large language models is crucial for enhancing the system's ability to recognize deontic modalities—obligations, permissions, prohibitions—within specific domain specifications. These models can be further trained to identify and interpret the nuanced usage of modal verbs in regulatory texts, which is vital for accurate deontic logical reasoning and automated rule generation. Furthermore, large language models simplify complexity through integration, enhance the efficiency of automated compliance checks, and reduce reliance on traditional NLP techniques. However, segmenting the text to fit the input constraints of BERT may lose contextual information and affect the accurate interpretation of complex regulatory text, and finer pre-processing is needed in future studies. Fifthly, enhancing the rule mapping methodology, such as the utilization of IFC files for semantic relationship mapping, is imperative. Additionally, our current research primarily utilizes Revit. Recognizing the need for broader applicability, future efforts will focus on extending our methodology to other BIM platforms, such as ArchiCAD and SketchUp. This expansion will enhance the scalability of our approach across various technological frameworks within the industry. Addressing these aspects constitutes the focus of future work in this area.

CRediT authorship contribution statement

Yian Chen: Writing – original draft, Supervision, Formal analysis, Data curation. **Huixian Jiang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision.

Declaration of competing interest

The authors declare no competing interests.

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Data availability

Data will be made available on request.

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